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An in-depth exploration of structural pose estimation strategies and datasets

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Abstract
Human Pose Estimation (HPE) refers to detecting the major joints (skeleton) of a human and their motion in order to recognize any action. This task is important because of its serious implications, such as surveillance, autonomous vehicles, sports analytics, human-computer applications, animation, and medical tracking. This article discusses various methods used to survey HPE, ranging from the fundamental principles of computer vision to advanced deep learning (DL) models. We investigate various structural pose estimation strategies for 2D and 3D domains and explore volumetric, planar, and kinematic models. Not only does this paper investigate methodology, but we also evaluate popular benchmark datasets used to test these methods, looking into their pros, cons, and overall usefulness for various HPE problems. Depth ambiguity, viewpoint variation, occlusion, and the absence of any annotated data are described as some of the many critical challenges to overcome. We further investigate what can be done to assist in the use of robotics and surveillance in real-time, such as self-supervised learning and adapting to specific task domains, aiming to drive future efforts forward.

Keywords Human pose estimation (HPE), Skeleton model, Joint localization, Pose tracking, Motion analysis, Surveillance, Human–computer interaction

1 Introduction

Pose estimation is a primary technique in computer vision that determines the configuration and location of limbs or body segments using sensor data in images or videos. By evaluating specific points representing joints and limbs, pose estimation aids in understanding both movement and body posture. To date, pose estimation technology has been investigated extensively owing to its applications in human-computer interaction, motion tracking, augmented reality (AR), virtual reality (VR), sports analytics, and medical diagnosis. Integrating structural information and motion data improves the precision and effectiveness of many gestures, behavior recognition, and rehabilitation intervention programs. However, pose estimation continues to be an open issue because of body posture variation, environmental occlusions, and changes in viewpoints. Additionally, while the models may be useful, real-world factors such as low light and partially visible people make that estimation less useful in unconstrained environments.

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The performance of DL pose estimation models continues to improve with progress in DL, surpassing the traditional OpenCV-based methods in object detection [1], semantic segmentation [2], and image classification. The advent of CNNs and transformers has greatly advanced pose estimation by allowing precise feature extraction and modeling of spatial relationships. Notwithstanding, DL-assisted pose estimation remains confronted with prominent issues, such as depth ambiguity problems, lack of sufficient labelled training data, and occlusions in a cluttered environment. These factors increase the difficulty of generalization across datasets, especially in the presence of occlusions and partially visible body parts. Training DL pose estimation models requires a lot of computing resources, making real-time use cases difficult to implement. As fully annotated datasets are also required, their creation proves difficult due to human error and the time needed to create such annotations. There have been attempts to lessen the dependence on labeled data through self-supervised and semi-supervised learning approaches. Still, these methods are unproven and require additional testing for actual use.

The estimation of poses is divided into 2D (two-dimensional) and 3D (three-dimensional). In pose estimation, 2D human poses are predicted from labeled two-dimensional images and videos. The introduction of DL has drastically improved accuracy in single-person cases. However, more complicated scenarios remain challenging, like multi-person poses estimation in occluded and crowded settings. On the other hand, 3D estimation of poses attempts to specify the coordinates of the body's joints in 3D space. This method captures human motion more completely. While motion capture devices can create very accurate 3D pose annotations within surrounding laboratory conditions, their use in the real world is usually limited because of hardware availability and environmental changes. Standard feature point detection models have limited accuracy in complex, unconstrained cases. This has driven the research toward hybrid techniques that combine depth cameras with inertial measurement units (IMUs). These can also improve pose estimation, enabling tracking in difficult scenarios where single-camera techniques cannot succeed. However, sophisticated hardware and calibration are required, which is costly and makes it difficult for such techniques to be extended to the public.

Pose estimation methods are broadly categorized as top-down and bottom-up approaches. In the top-down arrangement, a sole pose detector [3–5] first recognizes and crops each person in the image prior to their body part modeling [6–8]. These algorithms handle each recognized individual as a separate entity, so the neural networks can execute accurate joint detection using their powerful recognition capabilities. Top-down techniques are often more precise but more costly in computation since each person identified needs to be processed independently. Alternatively, bottom-up methods simultaneously find all the key points before performing the association. While easier for a multi-person case, bottom-up methods are burdened with key points grouping problems, which compromise their accuracy when clutter is high. Hence, these decisions should be made based on the needs of the intended application, such as the time frame, processing power available, or difficulty in the surrounding area. New developments in transformer-based architecture and graph convolutional networks (GCNs) have efficiently enhanced the accuracy of both top-down and bottom-up approaches, making it a favorable prospective research area.

Numerous studies and critiques have been written regarding pose estimation, addressing various issues. Some deal with vision-based human motion capture systems and include areas such as pose estimation, tracking, and activity recognition [9–11]. However, pose estimation is only one of the many subjects covered in these surveys. The authors of [12–14] attempt to trace the literature pertaining to three-dimensional (3D) pose estimation for research up to 2025, whereas [5] surveys single-view and multi-view pose estimation processes that are made through body-part decomposition. Between 2001 and 2024, most surveys concentrated on the more traditional physiognomic estimation systems instead of those based on DL methods. Survey [15] is the only one that considers most of the common DL approaches, though a small number of DL approaches offered have been analyzed. The analysis in [16] has been restricted to exclusively studying two-dimensional (2D) pose estimation techniques. At the same time [17], reviews monocular pose estimation from the prism of classical and DL methods, but it covers less than 100 publications up to 2020. Furthermore, it only tackles the issues of 2D single-frame monocular pose estimation and 3D single-frame monocular pose estimation. More difficult situations remain unexamined, such as 3D multi-view monocular video camera angle and 3D pose estimation through other sensor types.

Table 1 presents a synthesized classification of human pose estimation approaches drawn from existing literature, organized by feature types, motion cues, and body modeling techniques. Prior surveys have primarily focused on broad methodological taxonomies and high-level research directions, often neglecting detailed evaluations of specific

Table 1 A systematic classification of human pose estimation approaches

Components	Categories	Sub-categories
Features	Low-level features	(1) Shape: silhouettes [18–20], contours [21, 22], edges [23, 24]
		(2) Color: [17, 25, 26]
		(3) Appearance: [25]
	Mid-level features	(1) Local features: like Fourier descriptor [27], Shape Contexts [26], Dimensional Signature [28], Poisson Features [29], Histogram of Oriented Gradients (HOG) [30], Relational Edge Distribution [31], Scale Invariant Feature Transform (SIFT) [32], Edgelet Features [1], and Shapelet Features [30]
		(2) Global features: like Object Foreground Map [33], Max-Covering [27], Dense Grid Features [17]
		(3) like Hierarchical Model and X (HMAX) [34], Hyper Features [35], Spatial Pyramid [27], Vocabulary Tree and Multilevel Spatial Blocks (MSB) [36]
		(4) Automatic extracted features: like from a Convolutional Neural Network (CNN) [37–39]
	High-level features	(1) Full Context [40]
		(2) Combined human body parts [41, 42]
	Motion features	(1) Optical flow related: Dense Optical Flow [43], Robust Optical Flow [30]
		(2) Combined motion features, like Combined Edge Energy and Motion Boundaries [44]
Human body models	Kinematic models	(1) Pictorial Structure Model (PSM) [27, 45], Tree-Structured Model [27, 46], Improved Tree-Structured Model [27, 47]
		(2) Learned Pairwise Body Skeleton Part Relations [48–51], Learned Tree Structure Based on Bayesian Networks [25, 52]
	Planar	(1) Planar model: Active Shape Model (ASM) [25, 53], Cardboard [54]
	Volumetric models	(1) Cylindrical model [55]
		(2) Shape Completion and Animation of People (SCAPE) [56], Enhanced SCAPE Model [57], 3D Models with Shading [58], and Others [10, 59, 13, 60]

components critical to pose estimation. In particular, limited attention has been given to the role and evolution of Human Body Models, which are fundamental in capturing structural constraints and improving estimation accuracy. This survey addresses that gap by focusing specifically on Human Body Models used in posture estimation, offering an in-depth analysis of their design, underlying assumptions, and performance across various datasets. By concentrating on this core component, the paper provides a clearer understanding of how different modeling choices impact the effectiveness of human posture estimation systems in practical applications.

Although great strides have been made in pose estimation, some issues still need to be resolved, and further development in this area is still required. The practical implementation of pose estimation requires a model that can work under diverse conditions with respect to lighting, occlusions, and movements in the background. One of the most important shortcomings of current models is their lack of capability to transfer knowledge across different datasets without further adjustments. The combination of RGB, depth, and thermal modality adds to the model's robustness and generalizability; such integration can help solve many issues. Moreover, development in self-supervised and semi-supervised learning will help reduce the large dependency on pre-labelled datasets, making pose estimation models easier and simpler to implement. In addition, reconfiguring neural network designs for high processing speeds requires attention to implement pose estimation systems in real-time environments like automatic driving, monitoring, and virtual reality, which are time-sensitive challenges and place considerable strain on the models.

Extracting and representing pertinent features and key points through human body modeling techniques involves estimating a given pose and is a critical part of feature recognition [11, 27, 52, 58, 13]. These prominent key points are associated with highly relevant anatomical structures, like joints and limbs, that facilitate the form and motion of a person. Pose estimation frequently uses the N-joint rigid kinematic model representation, where the entire human body is abstracted as a set of joints connected by rigid links. This aids in incorporating computer algorithms that can compute the human body's movement, position, and orientation and is very useful in motion capture, activity recognition, and interfacing with the computer.

The human body is very sophisticated, with numerous details concerning its contour, kinematic system, and movement. The system is observed accurately for modelling body coordinates using sighted or model-based approaches with predefined structural representations. In pose estimation, the human body can be modeled mostly using three structural approaches: kinematic, planar, and volumetric. The kinematic model portrays the body as composed of interconnected bones and joints that allow flexible body movement, making motion detection easy. The planar model represents the body structurally using only two dimensions, as in images, to depict an object, which is useful in cases where depth is unnecessary. Finally, the volumetric model combines the other models as it considers the body's shape defined by its surface geometry and integrates the three dimensions to give a more realistic animation of motion synthesis and enhanced human body pose evaluation. These models contribute towards improving the systems in IR robotics and computer vision for robust pose estimation in different applications.

This document offers a thorough examination of the approaches, datasets, and analyses and poses estimation problems that have been previously documented. This work makes the following contributions:

- We examine different methodologies in structural pose estimation, both in the traditional computer vision-based and the recent DL-based paradigms. A discussion on the comparative benefits and drawbacks of top-down versus bottom-up approaches and their applicability in practice is presented.
- We identify publicly accessible datasets for the 2D and 3D spatial posture estimation and explain their features and relevance for various tasks as well as the processes of annotation. An evaluation of benchmark datasets is offered concerning their usefulness in harnessing the model performance and generalization capabilities.
- We focus on several pose estimation issues such as occlusion, depth ambiguity, adapting to different domains, insufficient annotated datasets, and real-time computing limits. Also, some possible solutions, like self-supervised learning, domain adaptation approaches, and multi-modal sensor fusion techniques, are discussed.
- We outline new directions for further research derived from the gaps identified in the current work—like occlusion handling using transformers, reduction of data dependency with self-supervision, real-time inference with lightweight models, and improved cross-dataset generalization from multi-modal sensors through domain adaptation.

This manuscript will be very useful for researchers who want to improve the field of pose estimation since it contains a thorough description of methodologies, datasets, challenges, and future scopes of this technology.

From a pose estimation research perspective, the paper is separated into five sections, each addressing a key area of concern in pose estimation. Section 1 provides an introduction followed by a literature review of pose estimation methods in 2D and 3D, such as kinematics, planar, and volumetric modeling in Sect. 2. This part also criticizes some computer vision techniques and deep modern learning approaches. Section 3 discusses the pose estimation-based datasets with an analysis of the most popular 2D and 3D benchmark datasets, their usage in model performance, evaluation metrics and methods of model performance evaluation and benchmarking. In Sect. 4, attention is given to problems and boundaries of the tasks in the pose estimation in the occlusion, depth ambiguity, and computing power problems. The section emphasizes the current research to address these problems with better modeling techniques, domain adaptation, and self-supervised learning. Finally, Sect. 5 speculates on future research directions focusing on the growing demand for efficient and robust real-time pose estimation techniques and the challenge of generalizing with multi-modal data. The manuscript captures the contemporary world and poses an overview of the field that will be instrumental to any formed or emerging researcher or practitioner in computer vision and artificial intelligence.

2 Recent trends in pose estimation

This chapter builds on the concepts laid out in Chap. 1 by delving into the history of pose estimation, especially recent memory of DL-based models, including modern architectures, modeling methods, and optimizations that shifted the understanding of datasets, development, challenges, and applications in later sections.

Pose estimation incorporates the latest multi-research area developments in robust accuracy, efficiency, reliability, and real-time responsiveness in DL. Other modalities such as RGB, depth, and inertial measurement have been multi-fused with the model, improving its performance in real-world scenarios. Multi-fusion, together with semi- and self-supervised models of learning, decreases the reliance on large sets of labelled data, increases performance over varying conditions, and generalizes many diverse variables and situations. Moreover, implementing lightweight models and sophisticated optimization techniques enables real-time inference for pose estimation, making it useful in healthcare robotics, augmented reality and human interaction automation. Further, researchers are working towards improving the handling of occlusions, viewpoint changes, and depth masking to improve the estimation in challenging scenarios. The pose estimation models can broadly be divided into kinematic planar and volumetric categories, as shown in Fig. 1. Animated models portray human figures as a joint hierarchy, allowing efficient motion tracking and skeletal analysis. The body structural planar models are simple 2D geometric models that are useful in applications where information is not useful, unlike volumetric models, which provide 3-dimensional detailed representations incorporating shape and surface geometry for synthesis of motion and detailed analysis of human body posture. Some state-of-the-art techniques based on those models are described below.

2.1 2D kinematic model

The growing integration of technology into everyday life has created a greater need for interaction between humans and machines, leading to significant advancements in artificial intelligence. Human activity recognition (HAR) is one of the most prominent subdivisions of AI, and it has been widely accepted, spanning health care, surveillance, and even human-computer interaction. It is now a fundamental area of research in AI. With attention toward 2D kinematic models, which focus on skeletal joint position analysis, a specific model is extracted, and isometric spatial filtering is conducted on two-dimensional images or videos. This approach has proven popular because of its computational efficiency and descriptiveness. These models use estimating postures to locate important

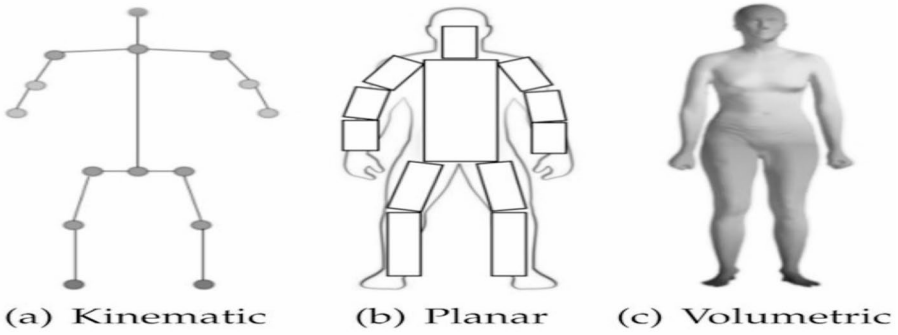


Fig. 1 Human body modeling approaches in pose estimation

joint markers, which allows the capture of spatio-temporal patterns needed for activity interpretation. The kinematics models are used because of occlusions and multi-person scenarios and the need for expensive 3D sensing devices. In addition, 2D skeletons or kinematics can be merged easily with heterogeneous data systems, improving the recognition level in sophisticated scenarios. With the maintenance of distinct skeletal frameworks and motion cues, working systems in real-time for 2D kinematic models are enabled for better usability in many scenarios.

In 2D Kinematic models, human orientation estimation has attracted a lot of attention in AI technologies during the last decades (pose estimation) [27, 61, 62]. The goal of pose estimation is to estimate the orientation of an individual skeletal body using sensor data. The research of pose estimation extends contemporary OpenCV employed in various domains such as HCI [11, 30], healthcare, motion analysis, VR and AR. Finding important segments of the human body skeleton from a given input image or video frame is called pose estimation. Creating novel DL approaches is also actively exploited in the 2D pose estimation research field. Besides, there are a variety of standard research procedures that employ artisanal feature extraction techniques.

Many forms of 2D kinematic models exist, such as single-person and multi-person human pose prediction. It aims to reconstruct 3D human poses using 2D human pose estimation data accurately and quickly from 2D pose detection (e.g., Cao et al. [64]) in multiple calibrated camera views. It mainly involves a multi-view matching or clustering problem that aims to uncover the correspondence between 2D observations seen in multiple views and the occurrence of false 2D results. By providing 2D observations as a collection of 2D skeleton recommendations with any number of components, we combine the advantages of both aspect and body association techniques (i.e., joint fusion in our approach). In Fig. 2, an overview of our method is given. To extract joints and their connections, we first input synchronized multi-view photos into a 2D pose detector (Fig. 2a, b). Next, we establish a skeleton space containing all possible skeletons, unlike part-level approaches directly connecting parts in 3D space. We organize each view's 2D joints and connections into a tree-structured graph to characterize the space for quick data access (Fig. 2c). Since each node in the network represents a joint and a node can have multiple child nodes for a particular child type, we can maintain all logical relationships between the identified joints. Finally, we reconstruct 3D poses by grouping

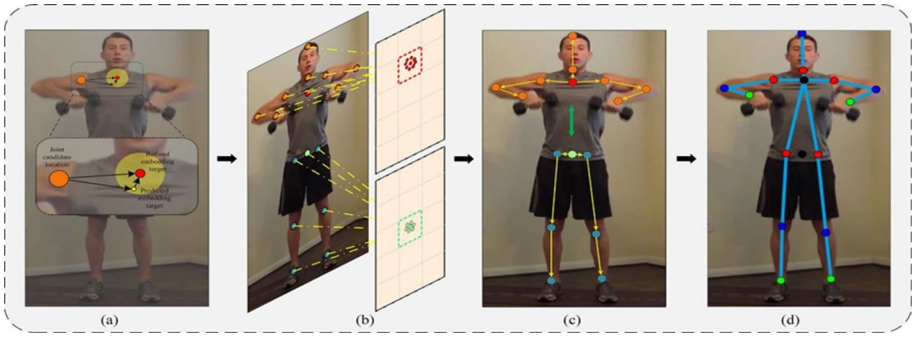


Fig. 2 Parallel greedy inference approach presented. **a** Calculating the human joint: the human joint location and its corresponding displaced heatmap values are employed to calculate the estimated integrating target, after which the anchoring joint's displaced heatmap is used to improve and obtain the enhanced embedding target. **b** Joint division. **c** Recommended parallel inference technique and sequence run within the DAEs-defined subspaces. **d** The overall result [63]

skeletons belonging to the same human from different views in space using a variant of the QuickShift approach, as shown in Fig. 2d.

2D kinematic models have become increasingly important in human action recognition (HAR) due to their ability to represent body posture and motion through skeletal joint coordinates. Recent studies have enhanced activity recognition by integrating these models with additional modalities like inertial sensors and RGB video [49] or leveraging deep learning architectures for improved spatial-temporal analysis [50]. Reviews have highlighted the efficiency and real-time capabilities of 2D pose-based systems [14, 51], while others focus on model interpretability [65] and practical deployment in surveillance applications [66]. These works demonstrate the versatility and continued advancement of 2D kinematic modeling, motivating further technical exploration in the following paragraphs.

In Li's [49] recent work, they evaluated a new framework that improves 2D skeletal-based HAR systems by combining heterogeneous sensor signals with skeletal representations obtained from 2D pose estimation. Their approach exploits data from IMUs, RGB video and even 2D skeletal joint coordinates to enhance activity representation. The model applies a joint embedding strategy to these diverse signals to harmonically merge them, thereby dealing with more complex recognition tasks, such as those found in real-world environments where multi-modal data is often noisy or incomplete. The framework is tested on multiple benchmarks involving 2D pose data and multi-sensor inputs. It achieves better results than previous models that used kinematic 2D measurements, especially in cases with overlapping or subtle motion sequences.

Yunardi et al. [50] did not investigate measuring inter-joint angle variations and joint swing amplitudes, more commonly associated with 2D skeletal kinematics. Instead, they focused on improving vision-based human action recognition and constructing a skeletal model for those actions. The motion patterns considered in their work can be characterized effectively in the 2D plane capturing spatial features. This research used a modified AlexNet model that included LSTM networks for capturing spatial and sequential temporal factors. Experimental results on some of the more complex surveillance datasets showcased greater improvements in accuracy recognition, especially in occlusion and low-resolution imaging scenarios. This method accentuates the value of integrating detailed 2D kinematic features with DL architectures on real-world problems to enable strong action recognition.

A comprehensive review of applying machine learning (ML) techniques to HAR was provided by Hossen and Abas [14], focusing on the developments in classical as well as DL models. The paper notes the increasing importance of approaches based on 2D kinematics that utilize skeletal joints for modeling activities due to their intuitiveness and efficiency. Other emerging trends include multi-modal data fusion, real-time HAR systems and attention mechanisms to enhance recognition accuracy. The authors also highlight major challenges, including occlusion, data sparsity, and computational resources in real-world applications. This work is vital for researchers looking to design robust and scalable HAR systems using 2D poses due to the insights it provides.

Krishna et al. [51] proposed a new framework for human action recognition using 2D posture analysis called MNCL, which is based on studying skeletal movement dynamics using Modified Neural Classification Logic. Their approach attempts to extract relevant features from the 2D skeletal joint data to account for the changes in human postures.

It utilizes a specialized neural classification structure, increasing the model's discrimination capability toward complex and similar actions. MNCL showed higher accuracy, robustness, and Prim's Acceptance Rate compared to other methods, particularly with real-time implementations, when tested on standard 2D pose benchmark datasets. This effort supports the development of 2D kinematic activity recognition systems that are efficient, interpretable, and aim toward simplifying design complexities.

Zhang and Wang [65] explored interpretable AI approaches for human action detection and recognition, which aim toward explainability without sacrificing accuracy. Their approach captures action dynamics temporally and spatially using 2D kinematic features obtained from human skeletal joints. The approach enhanced explainability while sustaining performance on recognition tasks, which positively impacted model trustworthiness. This research balances accuracy and interpretability in human-centric AI-implementing systems designed for real-world use. PosturalNet [66] uses OpenPose to analyze the joints of a person's body to detect shoplifting in large retail stores. It integrates feature encoding with dimensionality reduction to optimize efficiency while preserving accuracy. The synthetic and real surveillance data tests show that it has high detection accuracy (96.87%) as well as strong performance in occluded, real-world environments.

Despite their benefits, 2D kinematic models applied to HAR have several notable challenges and limitations. One primary challenge is the lack of a clear line of sight when capturing an individual in a frame along with an ambiguous estimate of a complete and meeting skeleton set projected onto the used image. Moreover, the absence of depth in 2D data further restricts the model's capabilities in capturing complex three-dimensional movements, making it hard to differentiate activities that may look the same in 2D but differ in spatial orientation. Because of occluded, interacting skeleton constructions, multi-person scenarios further complicate joint association and pose tracking. Furthermore, most existing algorithms for pose estimation tend to be excessively computational and require extensive annotated data or both, which limits applicability in real-time settings and scalability. At last, blending the 2D kinematic data with other sensor modalities remains an unsolved puzzle. The problem lies in different inputs that are not only homogeneous but also deal with the issues of occasional unsynchronized inputs and/or asynchronous time sequences. Tackling these obstacles is necessary to improve the consistency and functionality of 2D kinematic models in practical scenarios where the models will have to interact with real-world situations.

2.2 3D kinematic model

The development of Two-Dimensional HAR systems has heavily relied on two-dimensional kinematic models. These systems typically experience occlusions, view-dependent distortions, and depth ambiguities, which create problems from a singular viewpoint. Consequently, researchers have increasingly adopted three-dimensional (3D) models, offering better representations of human motion in space. The underlying reason for moving towards 3D modeling is the ability to capture the position of joints and movements of limbs in the three axes, which enables precise recognition of complex and subtle activities. Such accuracy is required in clinical gait analysis, virtual reality environments, intelligent surveillance, and biomechanics, where posture estimation and critical movement analysis are crucial.

The combination of inexpensive depth cameras, markerless motion capture systems, and advanced multi-view pose estimation techniques have accelerated the application of HAR Systems with 3D kinematic models. These models support the reliable recognition of actions across multiple scenes and subjects and minimize errors arising from occlusion or overlapping body parts. Also, 3D representations allow for more physically meaningful and semantically meaningful analyses of activities and enable advanced reasoning in context-aware AI systems.

The complete three-dimensional human skeleton is constructed with a RGB monocular video camera. The problem becomes very difficult when estimating 3D joint coordinates of a single view due to the uncertainty of depth and view. Nguyen et al. [67] proposed a two-stage framework intended to deal with such shortcomings. In the first stage, a CNN estimates 2D and 3D joint locations, specifically when a single frame capture fails to yield adequate results. The model is trained on a 3D human pose dataset and benefits from a 2D pose dataset, which improves spatial reasoning and generalization across various scenarios [68]. The second step uses a skeleton fitting procedure to the regressed locations of the joints and helps to ensure the pose reflects correct anatomy.

Kinematic Model Fitting and Skeleton Initialization [69, 70] are two essential components of 3D kinematic modeling used to estimate and track human poses in three-dimensional space. These processes transform raw joint predictions from DL models into accurate and temporally consistent skeletal representations that adhere to human anatomical constraints. They are critical for ensuring that 3D pose estimations are both spatially and physically plausible, especially when using monocular RGB video input, which lacks depth information.

Kinematic Model Fitting involves aligning predicted 2D and 3D joint coordinates with a predefined skeletal structure to form a coherent 3D pose. This fitting process incorporates spatial history from frame sequences and applies optimization techniques, such as least squares, to minimize discrepancies and maintain temporal smoothness. It also anchors the skeleton to the camera's coordinate space by fixing the pelvis joint. Skeleton Initialization, meanwhile, sets up an initial skeletal template that reflects the individual's proportions. This is done by averaging predictions over several frames and adjusting the template to match the subject's body ratios. Since monocular systems cannot estimate real-world scale, user-provided height is needed to calibrate the model. Together, these methods ensure precise, stable, and personalized 3D pose reconstruction across time [71].

Rani and Kumar [72] describe a smart surveillance system based on HAR with MobileNet and transfer learning, which shows efficient performance characteristics. Their methods provide high recognition accuracy while reducing computation costs, making them suitable for real-time systems on edge devices and mobile platforms. Improved detection performance is achieved by fine-tuning a pre-trained MobileNet on a human activity dataset while maintaining low complexity. This research illustrates the potential of Deployment-Ready HAR Systems based on compact DL models with transfer learning, which enable scalable and practical surveillance in real-time environments. Mathe et al. [73] address HAR occlusions with a new technique, generating artificially occluded samples to strengthen weak points within a given model through data augmentation. Their method works with the problem of occlusion in HAR by restricting visibility to certain parts of an involved human body during training. This simulation expands

the versatility of recognition systems under conditions where essential body joints are missing. It is observed that the application of the proposed approach on the benchmark dataset for HAR deteriorates the detection accuracy at the low level of occlusion but improves the detection accuracy at the higher level of occlusion, thus balancing the accuracy at different levels of occlusion. This study highlights the need for realistic data augmentation to develop robust HAR systems.

Meng et al. [74] propose a new technique for indoor HAR employing 3D LiDAR data. Their approach, Multiple Dynamic Nonlinear Mapping (MDNM), captures advanced spatial-temporal patterns from LiDAR point clouds. This method advances accuracy and robustness in comparison to traditional techniques. It efficiently deals with noise, occlusions, and diverse activity types. The results illustrate the effectiveness of 3D LiDAR with MDNM for privacy-sensitive, precise indoor activity recognition. Xin et al. [75] use 3D skeleton data to try and improve human action recognition recognition through DL techniques. They conduct a detailed study on different architectures of neural networks and their performance. The study underlines the impact of data augmentation techniques regarding the generalization capabilities of sML models. Results indicate that employing sophisticated DL models and effective augmentation improves recognition accuracy. This work is useful for developing reliable skeleton-based action recognition systems. Sharma et al. [76] propose Star-3D, a DL framework designed to recognize classroom human activities through 3D sensor data. It integrates 3D CNNs for spatiotemporal feature extraction, graph neural networks for interpersonal interaction modeling, and attention layers for dynamic feature learning. Occlusion effects and diverse postures are managed via preprocessing and data augmentation. Classroom experiments demonstrate that Star-3D significantly outperforms baseline models. This method improves recognition accuracy for activities in intricate, multi-person scenarios within indoor spaces.

3D kinematic models for HAR involve identifying and describing actions using skeletal joint movements and body poses. These models incorporate rich spatial and temporal information but have several challenges and shortcomings. First, accurate joint detection and tracking in real-life scenarios through occlusions, changing light levels, and sensor noise pose considerable difficulties due to incomplete or erroneous skeleton data. Second, high dimensionality and variability of human motions require expensive, complex, and prone models to overfitting if training data is limited. Third, diverse populations are often limited by subjects, body types, and movement styles, leading to a lack of robust adaptability. Less flexible constructions may incorrectly classify complex real-life activities, lowering recognition accuracy due to non-typical motion cues.

2.3 2D planar model

Interaction with a computer, monitoring of patients, surveillance, and even robotics are examples where estimating human pose has relevancy. Despite the diversity of approaches, the simple but sophisticated 2D planar model is vital concerning its ease of efficiency in computation. Unlike 3D models that require complex sensors, the planar approach works within the image plane as contours, silhouettes, and articulated segments are used, which lend themselves easily to monocular viewing. Defining these contours enables easy implementation in fitness applications and gesture recognition when paired with embedded systems. The utility of 2D planar models lies in their

practicality in real-world environments, especially where quick actuation and inexpensive processing are required. For example, mobile fitness apps, gesture recognition applications, animation pipelines, and educational aids rely on quick and lightweight 2D image processing models.

The computational efficiency and ease of application, especially in scenarios where depth information is not needed, makes the 2D planar model one of the most used methods in human pose estimation. Earlier planar models like the pictorial structure (PS) model [12], where the human body is represented as a set of rigid pieces joined by springs, have been very important. These models effectively detect poses in noisy conditions since they capture spatial relationships between body parts. This is made possible by considering body parts as spatially mobilized structures rather than a set of rigid segments. Later enhancements included deformable structures (DS) [77], where each body part is not defined as a rigid segment but as a deformable contour that can be represented with PCA using real images. Such representation improves the accuracy in depicting shapes of body parts subjected to different postures and points of view. Figure 3 illustrates the results of applying the 2D planner model.

Sturm's [78] pioneering work used homography techniques to enhance the pose estimation of planar objects using camera pose in conjunction with pose and geometric constraints, forming the basis of later developments in object tracking and augmented reality. Collins and Bartoli [79] built upon this using an infinite plane approach to pose estimation, improving speed and accuracy and enabling real-time operation in cluttered environments. These approaches are primary in unmanned motion tracking, surveillance, and gesture detection, and they tend to deal with minimal depth ambiguity and have limited processing power for 3D estimation.

In modern years, there has been a shift towards advanced 2D planar pose estimation using DL and generative techniques. One of the remarkable advancements is DiffusionPose [80], which proposes a novel human pose estimation framework by treating the task as a denoising process where the input is a keypoint heatmap viewed as noise, to be cleaned by a structure-guided diffusion decoder. This strategy is beneficial because it enforces more topological constraints on the skeleton, thus enhancing diffusion's occlusion and noise resilience. It outperformed existing models by a margin of 1.6% mAP on COCO and proved more accurate than earlier versions. Unlike earlier cope models that

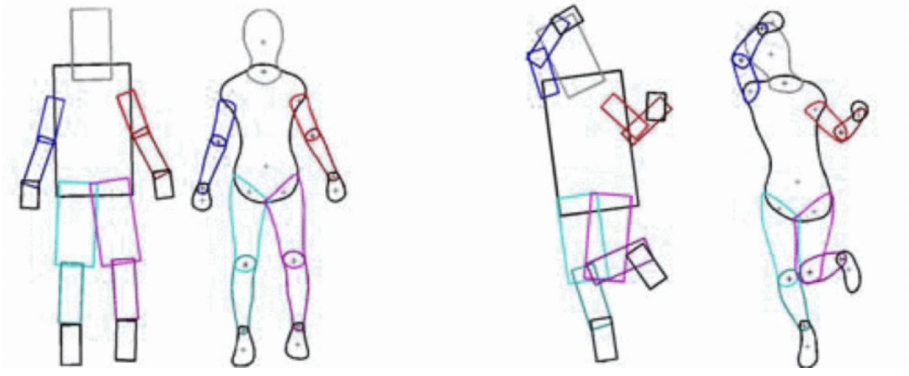


Fig. 3 Reconfigurable structures (the accurate model in every pair) are equivalent to PS modelling techniques in that they record 2D body shape deformations [12].

depended primarily on handcrafted part relations, this approach employs full spatial context via refinement throughout the denoising iterative process.

Another advancement is LiteDEKR, which modifies the DEKR framework for edged devices with low computational capabilities to make it lightweight [81]. This model uses LitePose instead of the costly HRNet backbone while adding Multi-Instance Bias Regression Loss and Object Keypoint Similarity (OKS)-based loss to facilitate full end-to-end training. LiteDEKR outperformed the previous models, suffering no losses in accuracy while cutting 75% of FLOPs and over 80% of parameters. This focus on optimising for lower mobile device and embedded system using real-time applications makes planar models useful in real-world situations like use in fitness apps and autonomous monitoring.

KSL-POSE [82] enhances pose estimation in real time by applying Kolmogorov–Arnold Network (KAN) modifications on YOLOv8-pose that add spline learnable activations increasing non-linear representation capabilities. Additionally, the model features a Small Object Enhance Pyramid (SOEP) and Layered Shared Convolution with Batch Normalization (LSCB) head to capture small keypoint features more effectively. These architectural changes allow KSL-POSE to surpass the baseline YOLOv8-l in precision and recall sprinting for COCO-POSE results, showcasing the adaptability of planar pose estimation models to speed and accuracy improvements simultaneously in dense, low-resolution environments.

The Transformer has also been an important development in providing solutions for 2D planar modeling. POT [83] is created to explicitly model the spatial relationships between the joints as a Pose-Oriented Self-Attention (PO-SA) mechanism works, encoding the distance between two joints while providing a layer of topology encapsulated within the attention maps. Also, it adds a distance-related position embedding to differentiate joint clusters relative to their anatomical hierarchy. For further increasing prediction accuracy, POT utilizes an Uncertainty-Guided Refinement Network (UGRN) that predicts the confidence of keypoint predictions and refines them selectively in areas of greatest uncertainty. These methods effectively achieve high pose accuracy for occluded or highly cluttered scenes, such as in the Human3.6 M and MPI-INF-3DHP benchmarks.

SkelFormer [84] provides a skeletal transformer architecture aimed at multi-view human pose estimation, marking a shift in focus. This design leverages inverse kinematics and poses priors by embedding them into the transformer framework. It enables the model to accurately convert noisy and incomplete 2D joint layout representations into unified pose and shape outputs. It is highly competent in occluded space and has demonstrated strong performance in and out distribution. SkelFormer is primarily a multi-view approach, but the strategies developed for structure-aware modeling remind us of the usefulness of 2D planar models when extended with complex transformer-based architectures. The study [85] explains how similar human activities can be recognized using millimeter-wave radar data. It applies CNN and Bidirectional Long Short-Term Memory network (BiLSTM) to capture spatial and temporal aspects. Class Activation Mapping is used to explain model decisions, which helps improve interpretability. The approach works well for activities that differ only slightly, demonstrating the promise of radar-based 2D planar models in HAR.

Recent advancements aside, existing 2D planar pose estimation models have significant gaps and shortcomings. One of the main issues is their capturing of depth, which creates problems when dealing with overlapping joints as poses become unduly complex or occluded. These models also struggle with pose estimation in intricate scenes. While DiffusionPose and SkelFormer have introduced newer models that improve robustness and accuracy, the dependence on large-scale annotated datasets diminishes generalization across diverse environments and subject variations. Additionally, performance in cluttered, low-resolution, or crowded settings becomes common, making real-time inference difficult when scaling these models for multi-person detection.

2.4 3D volumetric model

Human Activity Recognition (HAR) often requires voxel grids or occupancy data to capture full 3D volumetric human body models. These models represent the body and its motion over time using coarse-resolution 3D structures. Voxel representations are especially useful as they capture spatial information that 2D models cannot adequately represent. Unlike 2D volumetric models based on flat projections losing essential depth cues, 3D models accurately represent the body's posture and movements, ensuring resilience against changes in viewpoint or obstruction. This enhances recognition and analysis of complex actions and interactions, including subtle changes and interactivity with other people or objects. Affordable 3D sensors are now widely available, increasing the demand for these models in healthcare, robotics, and smart surveillance where understanding human activity is needed reliably and precisely.

In response to the growing interest in human three-dimensional remodeling, many volumetric body models have been published for various human body types. Although the parameterization of 3D body joints has been thoroughly researched [10, 59, 56, 58, 13], predicting 3D human form from a single shot remains a less focused and demanding topic. Previous research has advocated enhancing the spatial and structural aspects of 3D dynamic body modelling SCAPE [62, 64]. 3D form is modelled as a combination of location and structure factors in the SMPL [86] body model, which is still employed in more current techniques. Given a method and a source image, Bogo et al. [87] propose the SIMPLify modelling optimization technique. Rodin et al. [88] have impressively demonstrated human form retrieval from multi-view movies using optimization methods that use a sum of Gaussian volumetric models in combination with contour-based refinement. Even though such algorithms provide fascinating conclusions, they are ultimately constrained by the quality of the 2D detection employed. Furthermore, they apply priors on pose and form features to regularize a highly time-consuming and expensive optimization method. Figure 4 represents the possible outcomes of the 3D volumetric model.

The advantages of BodyNet are compared with other methods. BodyNet has 2D segmentation as well as 3D pose estimation. We define other methods by combining different combinations of the same components. SMPLify++. Lassner et al. [89] added a new term for 2D silhouette in their extension to SMPLify [90]. To enable a fair comparison with BodyNet, we extend it here. Using the algorithm of [67], we construct an acceptable target and include phrases for 2D silhouette and 3D stance in addition to 2D pose (see Appendix D). Even though both systems use 2D/3D pose and segmentation inputs, using MakeHuman [91], we generated a parametric skinned model and animated the

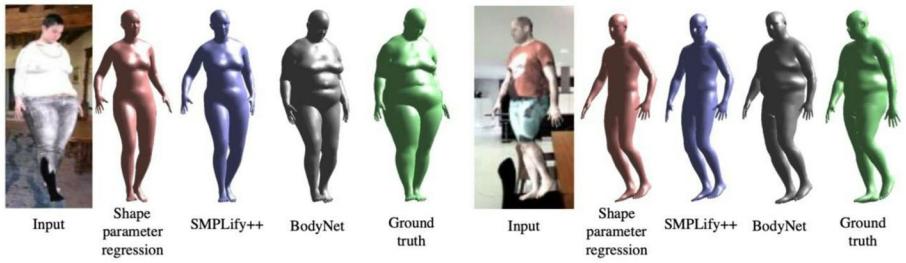


Fig. 4 Compared to other techniques, SMPL fits on BodyNet forecasts better. While shape variable regression and fitting simply to BodyNet key inputs (SM- PLify++) give comparable to average forms, BodyNet learns how the true shape seen in the image varies from the standard deformed shape models [86]

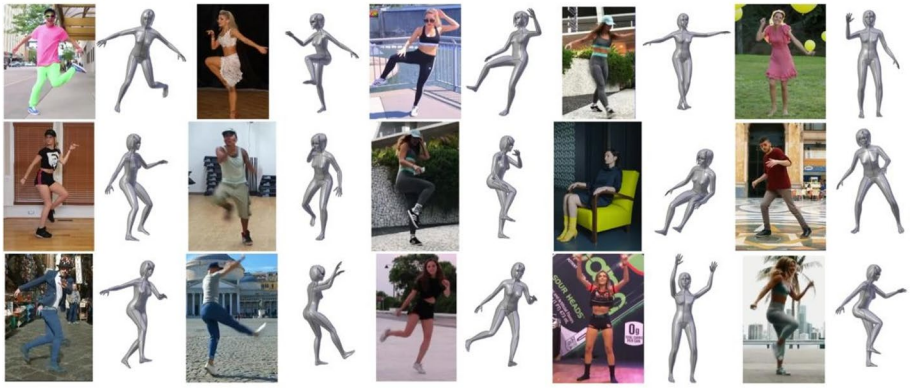


Fig. 5 MocapNET [93] qualitative results on YouTube dancing videos. Depiction of the computed coordinates using a skinned 3D human body modelling and samples of input RGB frames

armature using Blender [92]. Figure 5 provides samples of the results. The study's supporting material can provide many examples of output videos. Even if the joint location is wrong, you will often find that the motion quickly converges to the armature (this depends partly on the adjustable limb size of the individual model) (and on the variable limb size of the individual model). The error position usually results in a person standing too close to the camera, hiding his feet. This issue can be handled by having a class of experts perform joint estimation without the required feet. The second difficulty is the stance where (a) is standing.

Human action recognition integrated with Inception v3 for feature extraction from video frames at granular levels started with the work by Jayamohan and Yuvaraj [94] on the human action recognition framework. The model utilizes attention-augmented modified GRUs, outperforming traditional GRUs and enabling efficient capture of salient motion and temporal dependencies. This approach augments the spatiotemporal resistance of the dynamic's representation, thus alleviating long-term sequence modeling challenges in hybrid architectures. An infusion of convolutional neural networks (CNNs) into feature extraction using attentive recurrent neural networks achieves better results than other methods for benchmark datasets. This approach, which utilizes a deep learning feature hierarchy and context awareness refinement, improves the prediction of action in video sequences temporally.

A novel method of dynamic ground pressure attribute mapping for human activity analysis was proposed by Ray et al. [95]. Their model takes the pressure patterns in

various tasks and applies custom pressure features to enhance activity modeling. The system utilizes 3D simulated pressure data to model gait and posture, which improves the recognition and monitoring of physical behavior for healthcare applications. Their experimental proof of concept showed better flexibility and reliability than existing methods using dynamic recognition of pressure sensors.

Although 3D volumetric models provide depth spatio-temporal information by rendering human motion into volumes for HAR tasks, they still have some limitations and present challenges. The processing of dense voxel grids is computationally expensive and memory intensive and hinders scalability and real-time applications. A number of calibrated sensors, or high-quality depth data, usually result in accurate volumetric reconstruction but may not always exist in certain environments. Furthermore, the models tend to underperform with noise, occlusions, or missing data in dynamic or cluttered scenes, greatly lowering recognition accuracy. Ranging variations in human shape and size coupled with motion also make generalization harder across different subjects and activities. Lastly, the problem of crafting efficient feature extraction and learning algorithms for volumetric data remains an unsolved research problem.

Table 2 summarizes the differences regarding dimensionality, representation, advantages, disadvantages, and scope of application of 2D and 3D pose estimation models. 2D kinematic models are economically reasonable, which means they are lightweight and efficient simultaneously. However, their reliability in real-life situations is very low due to the absence of depth information. 3D kinematic models increase the accuracy with which an individual can be captured as they add depth, but they need more computation power. Skeletal 2D planar models enable shape tracking rather than joint tracking. 3D volumetric models maintain human realism to the greatest degree, but they are highly computationally expensive, making them ideal for animation, medical simulations, and virtual reality environments. The appropriate chosen model depends on the accuracy required in the task, computational complexity versus how much the task will be estimated, and the application for use.

Table 2 Comparative analysis of pose estimation models

Model type	Dimensions	Representation	Strengths	Weaknesses	Typical Uses
2D Kinematic	2D	Skeleton with connected joints	Lightweight, efficient; Good for gestures & motion in simple scenes	No depth; struggles with occlusions; limited 3D pose reconstruction.	HCI, Sports analysis; Surveillance; Motion capture
3D Kinematic	3D	3D joint-based skeleton	Depth information; Robust to occlusions; suitable for real-time tracking	Requires large datasets; Prone to errors from single views; Computationally demanding	Autonomous systems; Biomechanics; robotics; VR
2D Planar	2D	Deformable shapes & contours	Captures overall body shape; Good for silhouette-based tracking	Limited joint movement details; no depth; less accurate for action recognition.	AR; Animation; Fitness tracking; Behavior analysis
3D Volumetric	3D	Full 3D body shape & pose	Most accurate representation; realistic depth & geometry	Computationally expensive; Needs high-quality data; Slow optimization	Virtual humans; Gaming; Medical simulations; Cinematic animation.

3 Datasets and evaluations

In this chapter, we provide a detailed description of standard benchmark datasets and evaluation techniques for building and evaluating models for pose estimation, following previously discussed trends. This section highlights, through linking datasets to the previously discussed methods, how the multifaceted nature of data and thorough evaluation drives model performance and dependability.

Data sets are needed to perform pose estimation. They are also necessary to present a valid evaluation of alternative methods. This section presents the most widely used 2D and 3D datasets. The results of existing techniques are summarized in these major datasets. Figure 6 represents the hierarchy of the 2D and 3D datasets.

3.1 2D datasets

To develop and assess 2D pose estimation models, large-scale datasets with images of humans with annotated key points are imperative. These datasets contain many human activities, ranging in postures and intricate human interactions, which are critical to training models capable of dealing with real-world situations. Some datasets emphasize capturing everyday postures, while some focus on crowded scenes with multiple subjects in the same frame. Moreover, specific datasets for dynamic footed activities, such as dance or sports, enhance pose estimation accuracy with rapid and atypical movements. The key points in these datasets are major body segments, like the head, shoulders, elbows, wrists, hips, knees, and ankles that depict the human form as a skeleton. With large-scale provided images, researchers can test models for many criteria, such as performing occlusions, changes in viewpoint, or the level of complexity of the background with the model. These datasets are useful for fine-tuning pose estimation systems to increase accuracy and ensure their functionality in robust motion, human-computer interaction, and augmented reality systems. Some of the 2D datasets are as follows.

3.1.1 LSP (Leeds sports Pose) dataset

The Leeds Sports Pose (LSP) [96] is a popular benchmark dataset for human pose estimation, especially for sports-related activities. It contains 2,000 images, half meant for training and the other half for evaluation [97]. It is a great resource for pose estimation model development because all images represent a full-body pose, which is annotated with 14 key joints across the body. The images were harvested from Flickr and are sports and athletics-centric. It contains images of people performing badminton,

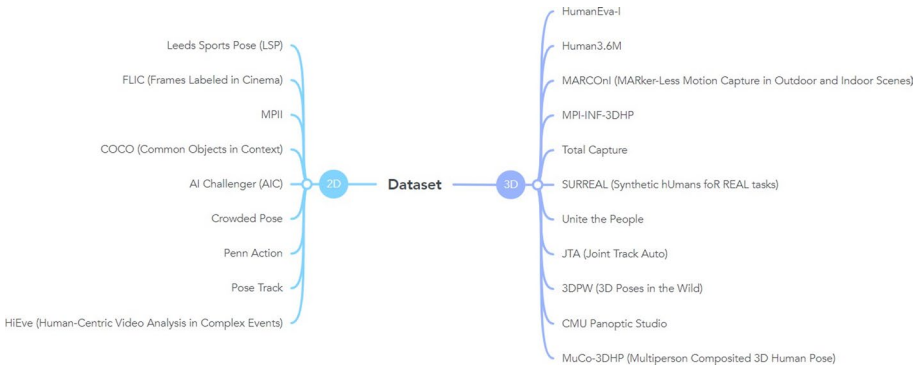


Fig. 6 Dataset representation

baseball, acrovers, running, football, squash, basketball, and other active movements. The variety of activities depicted in this dataset helps in building models that can perform difficult and multi-postured human poses in real life. To further improve the diversity and robustness of the dataset, an LSP augmented dataset was created, adding more than 10,000 additional training images to the original dataset. These images are taken from different backgrounds, poses, and angles to better cater for the generalization of DL models aimed at human pose estimation. This augmentation ensures that models trained on the LSP dataset can efficiently deal with occlusions, illumination, and various body poses. The LSP dataset and its augmented variations are widely applied in computer vision and DL, specifically in developing Keypoints detection, feature extraction, pose estimation and tracking algorithms. This dataset is used to improve models that recognize human body joints and motions for sophisticated systems in sports analytics, motion tracking, rehabilitation, and human-computer interaction. Some examples from the dataset are illustrated in Fig. 7, showing the variety of human poses in different types of sports.

3.1.2 FLIC dataset

The FLIC (Frames Labeled in Cinema) dataset [98] is instrumental in human upper body pose estimation and is captured from popular Hollywood movies. It contains 5000 frames, of which 4000 are allocated for training and 1000 for testing. The collection of human poses in the FLIC dataset is unique because it has been taken directly from movies, making it useful for researchers building human pose estimation models. An object detector was first applied to localize human candidates in the movie frames to build the FLIC dataset, which yielded around 20,000 sample detections. After detecting those candidates, an annotation procedure was conducted where 10 key joints of the upper body were marked. For maximum coverage in labelling FLIC, the labelled frames were sent to Amazon Mechanical Turk, where workers were paid to pinpoint the key points on the frames. With this method, a broad range of real-world upper-body poses were captured, increasing the use of the dataset in practical applications of pose estimations. By analyzing the frames, portions with participants that were not fully visible or were not directly looking at the camera were removed to create a specialized version of the dataset known as FLIC-Full. This improvement guarantees that the dataset has a high level of quality



Fig. 7 Lsp dataset [96]

with unambiguous and coherent upper-body pose annotations. The FLIC dataset and its full version have been extensively used in pose estimation computer vision and DL research to train and test the models offering human-computer interaction, activity recognition, and motion tracking functionalities. Some samples of the FLIC dataset demonstrating the variety of upper body poses captured in cinematic settings are illustrated in Fig. 8.

3.1.3 Human pose dataset MPII

The MPII Human Pose Dataset [99] is one of the most comprehensive and commonly used data collections for human pose estimation. It comprises around 40,000 recorded human poses obtained from real-life situations, 28,821 of which are training images, while 11,701 are test images. The database encompasses various human actions and interactions to validate pose estimation models. One notable feature of the MPII dataset is the variety of human actions and perspectives. For instance, the dataset includes people participating in leisure (e.g. walking, sitting, running), vocational (e.g. sporting, office work, and industrial work), and domestic (e.g. cooking, cleaning, and carrying objects) tasks. This great range of motions captured in this data set ensures that pose estimation models trained on this data set will perform well in generalizing across different contexts and activities. Every image in the dataset comes with meticulous annotations on the human's pose based on 15 joints, including the head, neck, shoulders, elbows, wrists, hips, knees, and ankles, along with associated occlusion features. The dataset al.so provides labels for occlusion, describing whether a given joint is fully visible, partially



Fig. 8 Flic-Full dataset [98]

occluded, or completely hidden. This feature makes it particularly useful for training models to perform real-world tasks that are very difficult, such as dealing with many people at once, as well as cluttered backgrounds and occlusions. The individual pose annotations in the MPII dataset are supplemented by activities designation and group poses for contextual purposes, enabling building models with more complex fictive estimation capabilities. This dataset is arguably the most recognized because of the range and variety of human postures, the diversity of viewpoint angles, and the complexity of the environment. Thus, the dataset has become a standard benchmark for DL for human pose estimation, action recognition and motion tracking towards developing sophisticated systems. Figure 9 presents examples of the use of occlusion and angle in the ease and variety of human activities captured in the dataset MPII.

3.1.4 COCO dataset

The Microsoft Common Objects in Context (COCO) dataset [100, 101] is one of the most popular, widely accepted and utilized benchmarks that are used for image computer vision. Its existence has contributed towards advances in research pertaining to object detection, image segmentation, key point estimation, and scene understanding. COCO is valuable for its diversity in pose, scale, occlusion, and multi-person scenarios, making it suitable for real-world applications in surveillance, AR/VR, and activity recognition. However, the absence of limb connectivity or motion information limits its direct application in biomechanical or motion-constrained modeling tasks. As a result, COCO supports graph-based keypoint models that are effective for joint localization but do not inherently represent a 2D kinematic structure. Despite this, the dataset is often used as a foundation for more structured models through post-processing or kinematic fitting techniques.

The dataset consists of approximately 200,000 high-resolution images with different real-life contexts, which are highly useful for training DL models for human pose estimation and object recognition. A prominent characteristic of the COCO dataset is its extensive and detailed annotations. It includes 250,000 key point-based human pose annotations that serve as a benchmark for 2D human pose estimation. The dataset comes with labeled key points for seventeen commonly found human joints, which include the head, shoulders, elbows, wrists, hips, knees, and ankles, among many other joints, along with multifarious positions and actions of humans in different environmental settings. These annotations are present in both the training and validation sets, helping researchers build and assess pose estimation models for pose estimation. In contrast,



Fig. 9 MPII dataset [99]

the test set lacks labels as it is reserved for benchmarking in competitive and problem-solving research endeavours. In addition to posing estimation, the COCO dataset allows for superpixel segmentation, instance segmentation, and dense image captioning, which makes it a valuable resource for multitasking learning in computer vision. Its diverse image set includes challenging scenes, such as indoor, outdoor, crowded, and occluded scenarios, ensuring COCO models perform well in real-world situations. The growing availability of the COCO dataset has facilitated its widespread adoption as the primary benchmark for estimating human location and pose from a single image in academia, industry, and AI-based automation. With its increasing scale and complexity, it remains a primary driver for developing DL models for autonomous systems, surveillance, augmented reality, and interactive AI systems. For example, human instances, their interactions with objects, and their intricate backgrounds on these datasets are verified from the COCO dataset illustrated and iterated in Fig. 10.

3.1.5 AI challenger (AIC) dataset

The AIC dataset [102], or AI Challenger dataset, is specifically designed for human key point detection, big feature data aggregation, and image annotation in Chinese labeling. It contains many images and annotations that allow for training and evaluating deep-learning models for human pose estimation and behaviour recognition. The Human Key Point Detection (HKD) subset is the most critical part of the AIC dataset, containing 300,000 images and 700,000 human instances with 14 key point annotations. These key points identify the head, shoulders, elbows, wrists, hips, knees, and ankles – essential portions of the human body. It is a perfect dataset to train pose estimation models. Unlike other datasets focusing on sports or controlled environments, AIC has images captured from common people's daily activities. This ensures that models trained on AIC can perform in many practical scenarios where human postures, occlusions, camera angles, lighting, or any combination of the above are challenging. One of the most notable features of the AIC dataset is its extensive multi-faceted feature data collection, which includes aggregation of pose data across different genders, settings, and social surroundings. Moreover, it is labelled in Chinese, which increases its value for AI applications localized to China and other Mandarin-speaking regions. The provided annotations within the Chinese labeling system greatly improve the quality of facial key points, body joints, and contextual metadata, thus making the data more useful in



Fig. 10 COCO dataset [101]

human-computer interaction, smart video surveillance systems, and AI motion analysis. The AIC dataset is widely used and is a benchmark for developing DL models in computer vision, action recognition, and intelligent robotics. Its relevance is also quite pronounced in human pose estimation, gesture recognition, and social behaviour analysis research. Because of its vast scope and diversity, it has become an important asset in intelligent surveillance systems, autonomous systems, and AI assistive technologies. Some of the different statistical perspectives capturing different postures, activities, and interactions of humans are shown in the examples of the AIC dataset in Fig. 11.

3.1.6 CrowdPose dataset

The CrowdPose dataset [103] is intended for human pose estimation in crowded places. Therefore, developing robust computer vision models to manage conclusions and overlapping people and human interactions is important. Unlike conventional pose estimation datasets, which focus on single and smaller groups of individuals, the CrowdPose offers a complex benchmark that includes multi-person scenario phenomena in which several people are close to each other within the same frame. Around 80,000 unique persons are marked over 20,000 high-resolution photographs comprising the dataset. The images were fetched from three publicly benchmark datasets based on a population factor that measures the number of people in the picture to guarantee that the dataset is balanced and representative. This population factor guarantees a variety of crowds within the dataset, from moderate use to very congested scenes, so that the pose estimation models could generalize effectively to different crowding levels.

The dataset is split into three subsets for training and evaluation purposes with a 5:1:4 split ratio:

- 50% (Training Set): Used to teach DL models to detect human poses in crowded environments.
- 10% (Validation Set): Used for model performance improvement and hyperparameter tuning.
- 40% (Test Set): The primary purpose was to determine the model's generalization and robustness in unfamiliar but busy settings.

The CrowdPose dataset is critical in ensemble learning of pose estimation, action recognition, and surveillance systems because it faces complex challenges such as occlusion, overlap of body parts, and person interactions. It is utilized to improve key point detection accuracy in dense crowds, multi-person tracking, AI-based crowds, public safety, and behaviour modelling automation. Sample images in Fig. 12 of the CrowdPose

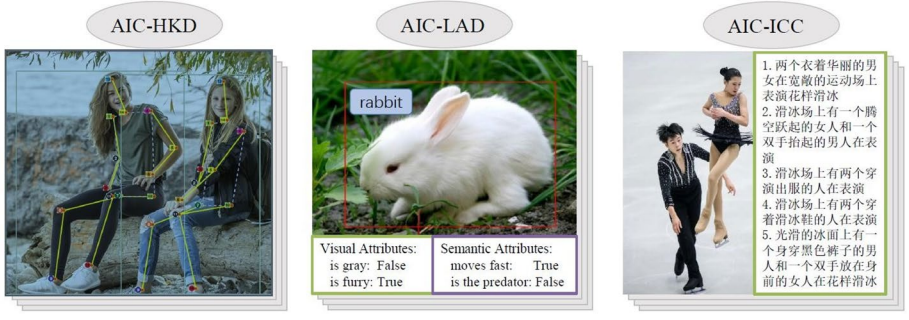


Fig. 11 AIC dataset [102]

dataset illustrate different crowd sizes, obstruction degrees, and body posture variations, demonstrating the dataset's complexity and realistic applications.

3.1.7 Penn action dataset

The Penn Action dataset [104] is one of the best human action recognition and pose estimation datasets. It was created to assess the estimation of human actions in real-life environments. Unlike image data sets, Penn Action comprises videos, making it useful for temporal pose tracking, limb movement anticipation, and human action recognition. This dataset comes with an enormous variety of human actions captured in various environments, thus aiding researchers in developing and perfecting deep-learning algorithms for various issues in computer vision. The dataset consists of 2,326 video clips, all of which are publicly available on YouTube, featuring 15 different actions such as baseball swing, basketball shot, bench press, bowling, golf swing, skipping rope, jumping, pull-up, push-up, sit-up, squat, tennis swing, throwing, walking, and kicking. These actions account for people's most common movements, which adds to their importance when training pose estimation and action recognition models. All the videos are optimally annotated, and a participant in the video has been identified using the 13 key points, which are the most important joints in the body, including a person's head, shoulders, elbows, wrists, hips, knees, and ankles. Furthermore, the dataset contains joint coordinate locations and visibility so that occlusion or motion blur does not hinder detecting and tracking the human body. To enable the development of robust models to be used in real-world action recognition tasks, the dataset is categorized into 1,258 training videos and 1,068 test videos for training and evaluation. The loose structure of Penn Action, which includes differences in camera angle, background clutter, and movement speed, makes it particularly challenging but also serves as an excellent benchmark for human activity understanding, sports analysis, and video surveillance activity detection. Sample frames depicting diverse human activities, ranging in complexity and annotation quality, from the Penn Action dataset are illustrated in Fig. 13. These diverse features show the importance of the dataset in improving action recognition and human pose estimation research.

3.1.8 Pose track dataset

The PoseTrack dataset [105] is a large-scale benchmark freely accessible by the public and intended for human pose estimation and articulated tracking in the wild. It is a relevant dataset to researchers focusing on multi-person pose tracking, motion analysis, and activity recognition. One of its distinguishing features is highly complex and dynamic regions where people undergo occlusion and close interaction, for instance, in public areas crowded with people. The dataset has evolved through different versions,



Fig. 12 CrowdPose dataset [103]



Fig. 13 Penn dataset [104]

with PoseTrack 2018 containing 1,138 video clips and 153,615 posture annotations. This makes PoseTrack 2018 one of the largest datasets for multi-frame pose tracking. Its earlier version, PoseTrack 2017, has 514 video segments with 16,219 human stance annotations, serving as a human motion analysis baseline. The dataset is not limited to single activities but captures a diverse range to ensure models trained on it generalize well to real life. To ensure proper documentation of movement and articulation patterns, the core movements for each of the thirty centre frames are well documented in the training videos. For testing videos or invalid videos, there are pose observations only every four frames so that researchers can analyze the continuity of motion and transitions between poses. The annotations incorporate fifteen articulations like head, shoulder, elbow, wrist, hip, knee, and ankle, which ensure that tracking of body movements is accurate and captured with no room for ambiguity. The PoseTrack dataset has proven immensely helpful when developing DL models for tracking human poses, performing real-time motion analytics, and activity recognition in videos. Its dense human interactions and complex occlusion scenarios make it a must-have dataset when working towards enhancing the accuracy and robustness of the pose estimation models for dynamic real-world conditions. Sample frames from the PoseTrack dataset are illustrated in sample figure frames with varying crowd density levels, occlusions, and human interactions. This emphasizes its importance as a testing benchmark for articulated pose tracking research. The instances of this paper are illustrated in Fig. 14.

3.1.9 HiEve dataset

The HiEye dataset, or HiEve [106], is a benchmark multi-person human pose estimation and behaviour recognition dataset provided through video recordings of people's actions to assist with human pose estimation, tracking and action recognition in challenging scenarios. It allows for assessing the effectiveness of DL models for multi-person pose estimation and behaviour recognition. The HiEve dataset contains 31 videos, each 7–31 s long, portraying various human actions and interactions. Each frame of those videos is accurately labeled, yielding over a million poses for humans where 14 joint locations were used for skeletonization. These joints, which form basic human postures,



Fig. 14 Pose track dataset [105]



Fig. 15 HiEve dataset [104]

include the head, shoulders, elbows, wrists, hips, knees, and ankles. The dataset has captured people in everyday situations such as office rooms, outdoors, public places, and crowded places, allowing accurate human tracking and movement forecasting.

The distinguishing feature of HiEve is that it focuses on three primary and highly human-centric tasks:

- Human Pose Estimation – highly accurate detection and localization of human body key points.
- Human Tracking – monitoring the same person through multiple video frames despite occlusions.
- Action Recognition – detection of human actions using physical postures and movements.

Fulfilling the requirements of highly detailed annotations and diverse action categories made HiEve an indispensable dataset for the progression of AI-powered human motion analysis, intelligent surveillance systems, and real-time behaviour modelling. It is also used to train DL-based pose estimation models for complex interactions, changing lighting conditions, and dealing with crowded environments. In Fig. 15, several instances from the HiEve dataset are shown, including several human poses, movement patterns, and representations of activities in the real world, which portray the dataset as challenging and rich for pose estimation and tracking research.

3.1.10 OCHuman dataset

The OCHuman dataset [107] is a challenging benchmark designed for advancing human pose estimation under severe occlusions. It contains 13,360 densely annotated human instances across 5,081 images, with an average occlusion rate of 57.3%, making it one of the most occlusion-heavy datasets available. Annotations include precise bounding boxes, 2D keypoint poses, and instance masks, enabling comprehensive evaluation of detection, pose estimation, and segmentation algorithms. The dataset focuses on crowded scenes where individuals are partially or fully obscured, posing significant challenges for conventional models. To improve robustness, techniques like "Occlusion Copy & Paste" augmentation have been developed using OCHuman, which synthetically introduces occlusions during training. The dataset and associated APIs available on GitHub facilitate reproducible research and the development of occlusion-resilient human analysis models.

Table 3 summarises widely used 2D human pose estimation datasets, highlighting their size, annotated key points, focus domains, special features, and typical applications. The datasets vary in complexity, from single-person sports poses (LSP, Penn Action) to crowded multi-person scenes with heavy occlusion (CrowdPose, OCHuman),

Table 3 Comparative analysis of 2D human pose datasets used in pose estimation research

Dataset	Size & type	Key points annotated	Focus / domain	Special features / notes	Typical applications
LSP (Leeds Sports Pose)	2,000 images (half train/test)	14 full-body joints	Sports and athletic poses	Augmented version adds 10,000 + images for diversity	Sports analytics, motion tracking, rehab, HCI
FLIC (Frames Labeled in Cinema)	5,000 frames (4,000 train, 1,000 test)	10 upper-body joints	Upper body, movie scenes	Annotated via Mechanical Turk, includes FLIC-Full for quality	Upper body pose estimation, activity recognition
MPII Human Pose	~40,000 images (28,821 train, 11,701 test)	15 full-body joints	Diverse real-life activities	Occlusion labeling, multi-context (leisure, work, domestic)	General pose estimation, occlusion handling
COCO (Common Objects in Context)	~200,000 images, 250,000 human instances	17 full-body joints	Everyday scenes, varied environments	Rich annotation, multi-tasking (segmentation, captioning)	Multi-task vision, pose estimation, AR
AI Challenger (AIC)	300,000 images, 700,000 human instances	14 full-body joints	Daily life, Chinese-labeled dataset	Large-scale, multi-faceted data, labeled in Chinese	Localized AI, surveillance, gesture recognition
CrowdPose	~20,000 images with 80,000 persons	14 full-body joints	Crowded scenes, multi-person poses	Population factor balancing, split into train/val/test	Crowd analysis, multi-person pose, surveillance
Penn Action	2,326 video clips	13 key joints (full body)	Action recognition, sports movements	Temporal info in videos, occlusion & visibility annotations	Action recognition, pose tracking, sports analysis
PoseTrack	1,138 videos (2018 version), 153,615 annotations	15 full-body joints	Multi-person poses tracking, video	Large-scale, multi-frame tracking, occlusion scenarios	Motion analysis, articulated pose tracking
HiEve	31 videos (7–31 s each), > 1 million poses	14 full-body joints	Human tracking, behaviour recognition	Focus on pose estimation, tracking, and action recognition	Real-time tracking, behavior modeling
OCHuman	5,081 images, 13,360 human instances	2D keypoint poses	Severe occlusions, crowded scenes	Average occlusion rate 57.3%, includes masks & bounding boxes	Occlusion-robust pose estimation

and include video-based tracking datasets (PoseTrack, HiEve). This overview aids in selecting appropriate datasets based on the target application and challenge focus.

3.2 3D datasets

3D datasets play a crucial role in training and evaluating models for 3D pose estimation. These datasets typically provide annotated 3D key points, depth information, and multi-view images, enabling models to learn spatial relationships and depth perception. They are widely used in applications like human motion tracking, robotics, and AR/VR. Some popular 3D pose estimation datasets include:

3.2.1 HumanEva-I

The HumanEva-I dataset [108] is a well-known human motion analysis and 3D pose estimation benchmark. It contains high-definition synchronized video sequences and three-dimensional coordinates of a human body's body position captured with motion capture systems. This dataset is vital in creating and testing DL algorithms for pose, motion, and activity recognition in two- and three-dimensional spaces. The HumanEva-I dataset consists of seven calibrated video clips: four in grayscale and three in color, all synchronized to 3D motion capture data. This calibration allows researchers to evaluate human movement from several angles, making the dataset appropriate for multi-view pose estimation and 3D reconstruction. The dataset aims to help design AI models to eliminate the gap between video-based human posture estimation systems and marker-based motion capture systems. The dataset has four participants performing six basic human actions, including walking, jogging, and gesturing. Each action has a video capturing different angles to ensure different movement styles like gait, posture, and limb movements. This difference increases the applicability of the dataset in domains such as sports analytics, robotics, and biomechanical engineering, where tracking human motion is highly critical. A significant benefit of the HumanEva-I dataset is that it has a very detailed level of annotation, as it contains both 2D and 3D key point spatial information, which enables the development and testing of DL techniques for human motion prediction and reconstruction models. This is especially useful for modern areas such as augmented reality (AR), virtual reality (VR), and intelligent surveillance systems. Figure 16 shows examples from the HumanEva-I dataset with images depicting each subject's actions. It also shows the multi-view and 3D pose annotations that make this dataset powerful for the analysis of human motion research.

3.2.2 Human3.6 M

The Human3.6 M dataset [109] is one of the most important and useful datasets concerning human motion capture and tracking, 3D action recognition, and human pose estimation. It sets a critical standard for DL models in computer vision, open-source robotics, and AI-based motion analysis due to the wide variety of human motion it captures and stores. This dataset includes around four million annotations of human poses with their joints accurately mapped in 3D space, together with high-definition video frames. Its sample capture frequency of 50 Hz allows for deeper, smoother motion tracking due to its high temporal resolution. The dataset's strength lies in its unique proposition of providing synchronized multimodal images, spanning RGB videos, depth videos, 3D skeletons, and IMU data, enabling AI upper body model training within diverse

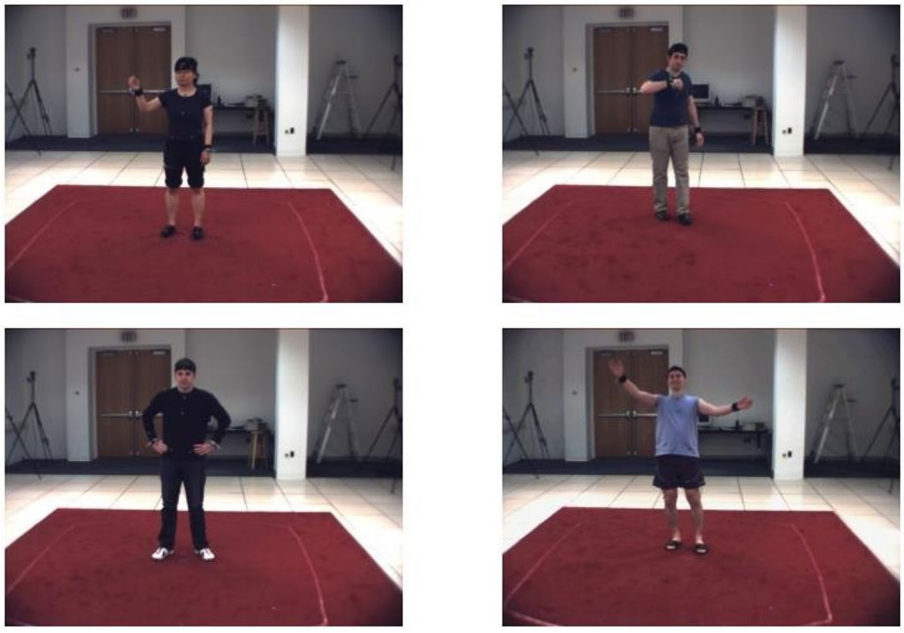


Fig. 16 HumanEva-I [108]

domains seamlessly. This dataset consists of seventeen real-world action scenarios performed by eleven professional actors. Some scenarios include talking to people, smoking, taking pictures, answering a phone call, sitting, walking, and eating. The dataset provides multi-viewpoint capture; these actions are recorded from four cameras. This is particularly useful for improving system generalization for human pose estimation and human behaviour analysis. A notable feature of the Human3.6 M dataset of human actions is the exceptional precision in the 3D localization of joints. Researchers can develop and validate algorithms for 3D human pose estimation, action detection, and activity forecasting thanks to the highly accurate motion capture system skeleton recordings. Moreover, the completeness of the dataset, combined with its variety of activities and multi-angle coverage, makes it very important for VR, AR, biomechanics, and advanced surveillance systems. Sample images from the Human3.6 M dataset are displayed in Fig. 17, which illustrates various action classes, multi-angle views, and comprehensive annotations of the 3D poses, which further demonstrates the role of the dataset, which serves as the benchmark for the assessment of motion tracking of humans in 3D space.

3.2.3 Marconi

The Marconi dataset [110] is a multi-faceted and broad dataset aimed at human motion analysis and activity recognition by enabling pose estimation. The dataset is rich in sequences needed to research various environmental sensor paradigms and motion complexities. It is useful for the training and testing AI human tracking systems and 3D motion capture models since the data is provided with numerous modalities for greater accuracy and robustness enhancement. The dataset is subdivided into twelve unique segments to capture human motion variations under different conditions. These included paradigm sensors, the type and number of cameras, methods of character identification, movement complexity level, and real-life scenarios. The dataset is captured with different types of camera systems like smartphones and GoPro video cameras, which are

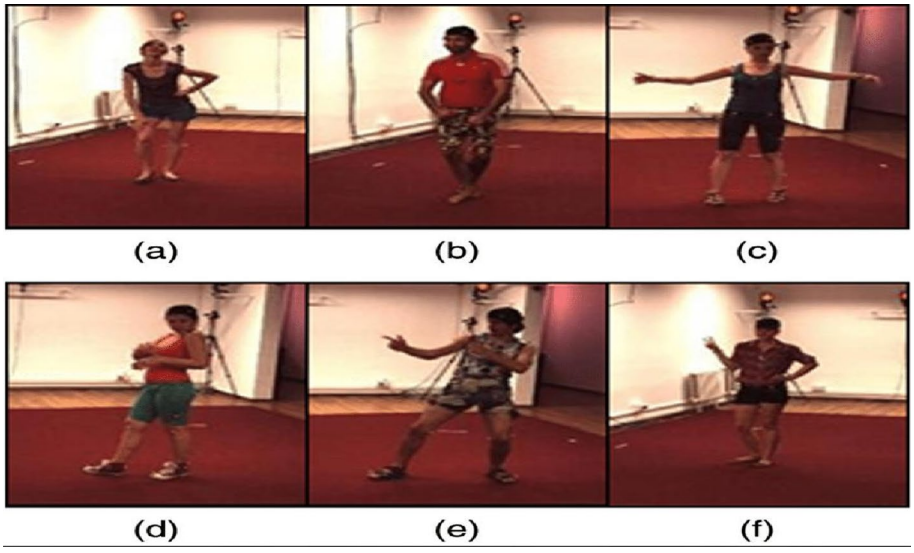


Fig. 17 Human3.6 M dataset [109]

time-synchronized for consistent multi-view recording. This time synchronization provides a diverse dataset for training models to work with different perspectives, cameras, and motion sequences. The Marconi dataset applies three distinct methods to obtain 3D coordinates of body joints and achieve remarkable accuracy in tracking motion and estimating poses. Those methods consist of:

- *Motion Capture System (MoCap)*: Some video sequences are captured with a Phase Space Active-LED marker-based system, which allows for accurate body motion tracking using infrared markers. This system utilizes these markers to provide high-accuracy 3D joint location data, essential for biomechanics and animation modelling.
- *A3D Method*: Besides other sequences, extra 2D poses are manually annotated to establish ground truth 3D joint locations. This method guarantees the accessibility of accurate 3D human pose data when the motion capture system is out of reach.
- *DMC (Dense Multi-Camera) Baseline Method*: In the cases where there are sufficient camera plants, the methodology involves extracting 3D point coordinates through multiliterate with several cameras.

This method uses stereoscopic vision to derive the position of a 3D human body from the combination of several 2D images taken from different cameras. The Marconi dataset is commonly applied to train DL models in human pose estimation, sports analysis, autonomy, and VR simulation. Its multi-modal features, rich motion activities, and synchronized cameras offer a treasure for researchers working on state-of-the-art human action analysis methods.

3.2.4 MPI-INF-3DHP

The MPI-INF-3DHP dataset [111] is a cutting-edge and multifaceted benchmark dataset used for 3D human pose estimation and motion capture. Unlike classical motion capture databases that depend on wearing specialized suits or reflective markers, MPI-INF-3DHP uses a professional annotation-based motion capture system that permits the actors to wear casual everyday clothing, such as loose clothing. This increases the applicability of the dataset to real-life scenarios where those restrictions are absent, which

enhances the generalization capabilities of DL models trained on it. The dataset contains eight participants: four males and four females. Each participant performs eight action sequences, each lasting roughly one minute. These sequences include walking, sitting, complex training exercises, and other highly flexible and coordinated movements. The dataset is more versatile in motion classes than the Human3.6 M dataset, which makes the dataset more useful for training AI in pose estimation and movement recognition as well as in sports. The testing portion of the MPI-INF-3DHP dataset consists of 2,929 valid frames extracted from six different users performing seven activities. One of the most notable features of this dataset is that it contains both simple and patterned sets of clothing, which increases clothes' variability a lot more. This feature guarantees that AI models trained using this dataset could accurately estimate human poses for different styles, textures, and complexities of motion combined with clothing. Including chroma-key masks for background removal and enhancement is another unique aspect of MPI-INF-3DHP. This aids in separating the foreground from the background for better scene augmentation, background replacement, and environment adaptation to pose estimation tasks. The dataset is highly flexible for AR and VR systems and real-time motion analysis because it allows modifications to both the foreground and background. Sample frames from the MPI-INF-3DHP dataset are presented in Fig. 18 and depict various action sequences, clothing styles, and foreground and background segmentation for each action performed, which turns the dataset into a strong and flexible benchmark for 3D human pose estimation research.

3.2.5 Total capture

For analyzing human activities, actions, pose estimation, or motion in detail, the TotalCapture dataset [112] is the first multi-modal dataset. The dataset is augmented by skeletal annotations collected using multi-view video, inertial measurement units, and a commercial-grade motion capture system, Vicon, so it is the first motion capture dataset to use this level of detail for video understanding. The outline of vision-based inertial and motion capture data with such detail makes the dataset one of the most comprehensive, diverse, and rich for human motion research. Instead of the conventional



Fig. 18 MPI-INF-3DHP dataset [111]

TotalCapture motion capture datasets, which require stylized special suits or reflector markers, performers clothed in open ordinary clothing without any stylization are affordable. This maximizes the visual diversity that total capture seeks, and therefore, the dataset enables human motion capture in everyday realistic settings. The dataset is imaged with careful tracking for Ankle, Head, Upper, Lower Back, Upper, Lower Arm, and Foot position Data to enable easy HAR. The placement of these sensors is expected to improve biomechanics and movement dynamics, capturing accuracy. Unlike other Capture datasets, TotalCapture offers unique primitive pixel-wise background subtraction, naturally enabling humans and motion segmentation. Such a feature ensures that models estimated using this dataset can eliminate noise unless they aggregate human motion, allowing augmented reality (AR), virtual reality (VR), and markerless motion capture systems to function optimally without hitches. Every user in the dataset performed the five activities three times, each with various movements. This ensures that TotalCapture can still be used as a comprehensive reference for DL models that aim at pose estimation, motion tracking, or action classification. Additionally, the dataset is divided into subclasses corresponding to object and action sequences to enable focused investigations on specific motion patterns and behavioural interactions. Snapshots from the TotalCapture dataset are presented in Fig. 19, showcasing its multi-view configuration, placement of the IMU sensors, various motion patterns, and accurate human motion capturing features, which establish TotalCapture as one of the primary datasets in the field of human motion analysis.

3.2.6 SURREAL (synthetic humans for REAL tasks)

SURREAL (Synthetic Humans for Real Tasks) dataset [113] aims to unify four advanced tasks in one dataset, making it easy to perform 3D human body posture estimation alongside shape analysis and action recognition. This dataset has been generated to facilitate research in DL for motion analysis and computer vision. It trains models with images, textures, shapes, human bodies, and environments. The geometry information for SURREAL comes from the well-known CMU Motion Capture (MoCap) Collection, a repository for capturing anthropomorphic motion. With this motion data, realistic human body shapes in 3D are constructed using the Skinned Multi-Person Linear (SMPL) model. These synthetic human models are crafted using specific known parameters to make them look like real humans. The SMPL model provides enhanced body deformation control; thus, multi-view pose estimation, human tracking, and motion



Fig. 19 Total capture instances

prediction become easier with SURREAL. SURREAL differs from other datasets' texture generation features, making synthetically generated humans more realistic. To enhance the quality and diversity of the synthetic training data, the context includes textural information gathered from the 3D scans of people in casual outfits. This guarantees that the artificial humans in SURREAL are dressed in different clothing styles, have realistic skin textures, and their body movements are natural, thus improving the representativity of the models to the real world. SURREAL uses a 9-coefficient spherical harmonics model for lighting simulation enabling good shadowing and illumination. With this sophisticated model, normal human figures can be rendered to interact with virtual settings and enhance the usefulness of the dataset in fields such as computer vision, gaming, animation, and augmented reality (AR). For the first time, SURREAL, as a dataset, implements 3D pose annotations, body parts segmentation, and optical flow ground truth for multitask training. Moreover, the dataset has posed estimating features, depth maps, and body part segmentation with the ability to analyze human motion, body morphology, and spatial relations. The dataset has 3.6 million frames, each of which is labelled, making it one of the largest datasets of annotated artificial humans for research purposes. To maintain an orderly learning progression, the data was split into a training set and a testing set, which included 115 people in the training set and 30 people in the test set. These measures guarantee that models trained on SURREAL can generalize to new subjects and unseen movements. Sample cases from the SURREAL dataset showing different human poses are annotated with synthetic body shapes, textures, and multimodal annotations, which are depicted in Fig. 20, further proving the dataset's importance as an initial dataset for 3D human motion analysis.

3.2.7 Unite the people

The Unite the People dataset [89] assists in 3D human body modeling, pose estimation, and shape prediction. It has 6,777 images, consisting of 5,569 training images and 1,208 test images, which aid in DL-based human pose and shape estimation. Some sample instances from the dataset are illustrated in Fig. 21. The dataset was created because CNNs are usually taught using separate, isolated datasets like MPII and LSP, which lack 3D body prediction components. To fill this void, the authors improved the SMPLify technique for creating high-resolution 3D body models and then handpicked the best-fitting models to form a multitude of human pose datasets. This union allowed them to merge human body representations from various datasets, thus facilitating training models equipped to perform better on many data sources. A semi-automated method was employed to improve labelling precision and correct the annotation. By reprojecting

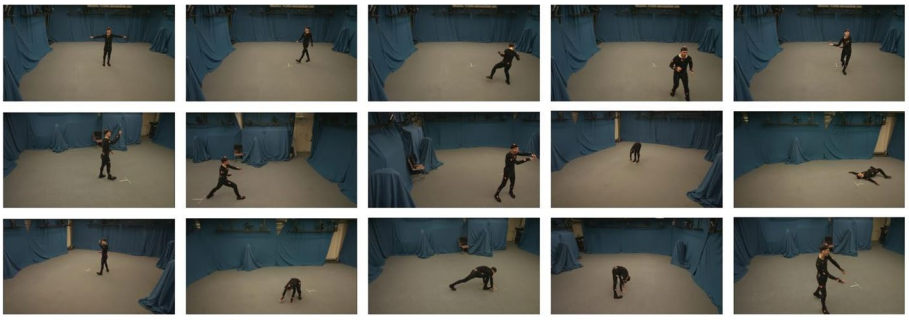


Fig. 20 Surreal dataset [113]

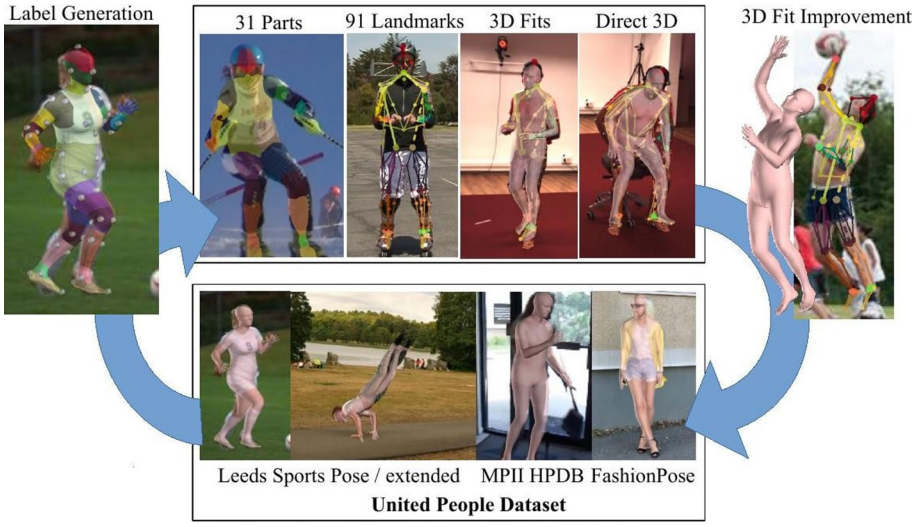


Fig. 21 Unite people dataset

the actual 3D body model back onto the original image, this technique accurately matched image-based poses to their corresponding 3D representations. This effective labeling system enhances the quality of human dataset annotations without losing integrity throughout the datasets formed from different humans. One of the key advantages of Unite the People is its dense set of labels, such as the annotations for 31 distinct body parts at level 1 and 91 distinct landmark locations at level 2. The rich annotation framework greatly mitigates the issues of human pose and shape estimation uncertainty, even in conditions with challenging occlusions, body shape variation, and different poses owing to the variation in the contour. The dataset enables the model with the detailed label description to train itself properly and enhance the efficiency in figuring out the imprecise human pose and shape along the torso. Besides, the authors present a regression tree model [165] for posture and shape prediction, which is one or two orders of magnitudes SMPLify slower, hence less computationally complex. With this, the computational intensity is significantly reduced, enabling fast 3D pose estimation for real-time applications like animation, virtual reality (VR), human-computer interactive systems, and biomechanics. Ultimately, experimental research suggests that pose estimation models can be trained with minimal data using 91 body landmarks, which does away with gender-based or positional constraints. This makes Unite the People a highly versatile and non-discriminative dataset, which can be effectively used for human pose estimation in AI-assisted healthcare, motion analysis, and synthetic modelling of humans, proving the dataset's utility in arms without discrimination.

3.2.8 JTA (joint track auto)

The JTA (joint track auto) dataset [114] comprises synthetic human surveillance for human tracking, multi-person pose estimation, and action recognition tasks in the city. Diversifying from traditional datasets that relied on real-life videos, JTA takes advantage of the computer-generated scenes from the Grand Theft Auto V (GTA V) developed by Rockstar North. This dataset offers the ideal environment for DL models with accurate 3D pose annotations, ground truth classifications, multi-angle views and captures of the model. The dataset has 256 training clips and 256 test clips, all with

multi-angle, multi-light conditions to enhance realism. It has one of the widest capture ranges of viewpoints, lighting, and object interactions. It introduces the dataset as a realistic benchmark for pose estimation and tracking in crowded urban conditions. The JTA dataset includes the capture of movements of pedestrians in dense traffic, at intersections, on the sidewalks, and in open spaces, which ensures a variety of poses and interactions. JTA stands out due to its realism in in-depth variation of the machine to target distance. From 0.1 to 100 m, the data range is useful for models to accurately track humans from both close and far views, which is helpful for use case scenarios in autonomous driving, intelligent surveillance, and crowd behavior analysis. Further, the dataset's object height ranges from 20 pixels to 1,100 pixels, which provides varying resolutions for testing model performance across different scales and resolutions. The dataset contains 14 automatically annotated human body components with the game's rendering engine, allowing for precise 3D pose annotations. This automation makes JTA one of the most accurate synthetic datasets for pose estimation. In addition, the dataset incorporates some real-world obstacles, such as occlusion and self-occlusion, which make pedestrian tracking more challenging: Occlusion. In this circumstance, an external object such as a pedestrian, vehicle, or other structure blocks the view of a body joint. Self-Occlusion – A phenomenon whereby key joints are obscured because of one's body parts, making it difficult to estimate the position and posture of the hidden limbs. In addition to pose estimation, JTA also includes individual re-identification tracking, enabling unique identifiers for each pedestrian for identity management within video sequences. Such information is indispensable for behaviour analysis, anomaly detection, and security monitoring, which rely heavily on identity-tracking features. With game-rendered environments, JTA has developed a new dataset that is more cost-effective and easier to scale compared to others, making it possible for scientists to test AI-based human motion tracking in sophisticated urban environments without the costs associated with traditional datasets. Figure 22 displays sample frames from the JTA dataset, revealing many urban scenes with multi-scale views, lighting changes, and occlusions. Such scenes make this dataset useful for contemporary pedestrian tracking and pose estimation research using AI, as demonstrated by the variety of resources available within the JTA.

3.2.9 3DPW

The 3DPW (3D Poses in the Wild) dataset [114] is one of the first in its field and provides a set of 3D human pose annotations tailored towards real-world motions. Unlike many laboratory-based datasets, 3DPW was created to facilitate appeal assessments in



Fig. 22 Joint track auto dataset [114]

practical settings. It represents a robust benchmark dataset for image-based 3D human pose estimation and motion tracking. The dataset is constructed by merging data from Inertial Measurement Units (IMUs) with that of a phone camera worn by the user. This configuration provides accurate pose estimation in static and dynamic scenarios, allowing natural human motions to be captured while performing different activities. To create precise 3D human posture data, 3DPW initially utilizes a technique called Video Inertial Poser (VIP) for three-dimensional pose estimation. The technique is based on merging image frames from video clips with IMU sensor information, where each frame comes with a label containing real-time 3D pose information. VIP method is validated with the Whole Acquisition dataset and achieves a high estimated position accuracy of 26 mm, which is favorable for building DL models aimed at 3D pose estimation. The dataset includes various human activities and movements with corresponding pose annotations from the real world. It contains videos of people performing daily routines, including walking around the city, climbing stairs, drinking coffee, and driving a bus. It also includes up to 18 outfits per individual, ensuring different clothing styles and textures. It is important for training the models to be invariant to clothing changes that affect the pose estimation. To monitor human motion more accurately, 3DPW employs 17 IMUs per subject, whereas 2 to 3 subjects can be tracked using 9 to 10 IMUs. The placement of the IMU sensors on the major bodily parts guarantees suitable skeletal tracking and body posture recognition. Moreover, a clap was used as a synchronization movement for video recording and IMU data so that the motion sequences matched the corresponding IMU outputs perfectly. The 3DPW dataset is an outstanding resource for DL computer vision tasks that involve humans, particularly body part tracking, biomechanics, activity detection, and even augmented reality systems. Realistic and dynamic human motion sequences make it one of the most useful datasets for training AI-based 3D human pose estimation systems in practical environments. Figure 23 displays sample frames from the 3DPW dataset, depicting different human activities, multiple environmental conditions, and integrated motion capture, underlining its relevance in human-centred artificial intelligence and 3D pose estimation progress.

3.2.10 CMU Panoptic Studios

The CMU Panoptic Studio dataset [115] is considered among the best in multi-view human motion capture, social interaction analysis, and 3D human pose estimation [116].

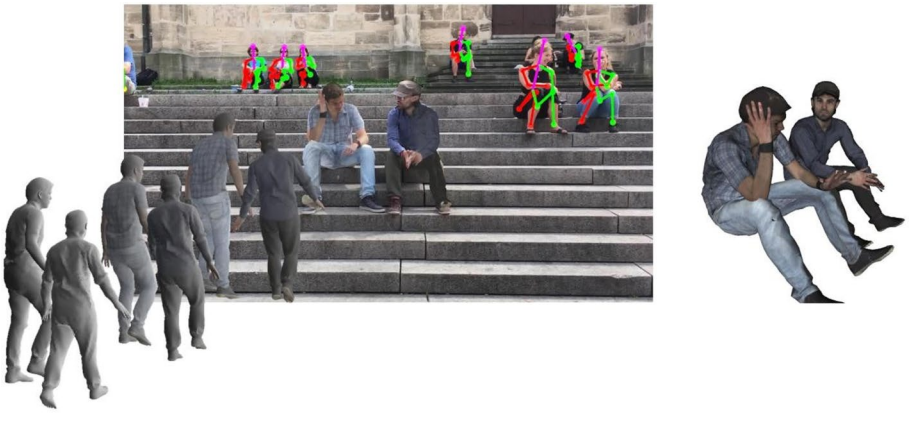


Fig. 23 3DPW dataset [114]

Tailored for capturing large-scale, high-resolution human activity data, it sets the mark for DL frameworks for human tracking, behavior recognition, and motion prediction. This dataset's primary advantage is its large multi-camera system, which contains 480 synchronized VGA cameras capturing high-density representations of human movement from various angles. 30 synchronized HD cameras offering temporally aligned high-resolution data with the VGA cameras. 10 RGB-D sensors allow depth perception and estimation of 3D human poses in real-time scenarios. This vast multi-camera network enables accurate 3D multi-person motion tracking and the reconstruction of human actions, thus setting the gold standard for AI-powered pose estimation, action recognition, and social activity recognition. In the CMU Panoptic studio dataset, the motion calibration steps maintain the synchronization and alignment of the 521 cameras (480 VGA, 30 HD, 10 RGB-D), thus reducing the errors associated with pose estimation and skeletal tracking. The dataset simultaneously records human behaviours, social activities, and group interactions, which are imperative for interaction analysis in human-computer interaction, multi-person tracking, and crowd behaviour evaluation. One of the most important aspects of the dataset is the ability to evaluate social interactions in real-world settings. Unlike most traditional motion capture datasets where subjects are individuals, the CMU Panoptic Studio dataset focuses on group behaviours such as mobile joint attention and multi-person interactions, as shown in Fig. 24. The dataset is exceptionally suitable for AI applications in robotics, surveillance, virtual reality (VR), and biomechanics due to the high-quality 3D motion annotations coupled with multi-view setup and synchronized depth sensors. The anatomy of human movements, gestures, and social actions is made possible with the aid of the dataset, which greatly supports the development of computer vision and AI-driven motion analysis.

The development of 3D motion data sets has greatly aided human action recognition, pose estimation, and motion analysis. Initial data sets were based on indoor multi-view videos with marker less motion capture, but they widened in scope and variety with time. Datasets gradually added multiple subjects, more elaborate motion sequences, and added augmentation data to improve robustness. Some datasets focused on real-life scenarios, whereas others added synthetic data and segmentation features. Further developments incorporated IMU and motion capture systems and developed more precise annotating strategies alongside large-scale simulated datasets. The posed challenge was successfully solved by the accumulating difficulty and variety of data sets, which



Fig. 24 CMU Panoptic dataset

enhanced the accuracy and generalization of the 3D pose estimation models. The variety of datasets is compared in Table 4.

3.3 Evaluation metrics

It is essential to have clear metrics [5, 21, 117, 101, 39, 70, 118] that evaluate the accuracy, robustness, and overall effectiveness of pose estimation models to evaluate them appropriately. Looking at 2D pose estimation, which is concerned with key point localization in an image, is much different than looking at 3D pose estimation, which incorporates predicting joint positions in space. The effectiveness of these models needs to be measured by real-world applicable metrics, which serve as the ideal comparison means for models.

One of the most common indicators for 2D pose estimation is the "Percentage of Correct Key Points" or "PCK." This indicator pertains to the ratio of expected key points correctly estimated within a certain distance of the true value. The distance ranges are set as a percentage of certain body part lengths, such as a head or torso, to make them adjustable to different scales. One variation of this measure, called PCKh, adjusts the range based on the size of the subject's head, facilitating greater accuracy when dealing with different body proportions. Another metric is Object points similarity or OKS, defined in the context of the COCO Pose Estimation Challenge, which is also widely used. OKS is defined and used in the same manner as Intersection over Union (IoU) in object detection, and it estimates the accuracy of the localization of the reference key points as a Gaussian-weighted score of the predicted and the expected key points squared. Furthermore, an important asset to assist mAp is derived from the COCO challenge, which estimates model predictions with different thresholds of OKS measured at the same time of recall and precision. The PCK is just one method to assess the accuracy of 2D pose estimation; the other one is NME or Normalized Mean Error. This measures the Euclidian metric distance between the predicted key points and the expected key points set within a certain range defined by a scale such as inter-ocular distance.

The evaluation of the models is performed according to the accuracy of predicting the joints of the object in 3D space in real-world scenarios in 3D pose estimation. The most common measurement for this is the Mean Per Joint Position Error (MPJPE), which measures the mean of the Euclidean distances between the estimated 3D joint positions and their corresponding actual positions. However, the MPJPE always suffers from global misalignment errors, which makes it unsuitable for cross-dataset evaluations. For this purpose, Procrustes-Aligned MPJPE (PA-MPJPE) modification computed the same error after applying a rigid body transformation to the predicted and ground-truth poses, thus removing any non-structural differences in aligned poses. Another evaluation metric that deserves attention is the Mean Per Joint Velocity Error (MPJVE), which qualitatively assesses the temporal coherence of predicted poses by the difference of joint movements in time and is assumed to be a relevant motion. This metric is crucial for measuring the smoothness and realism of the motion for video-based pose estimation tasks. The extension of the PCK metric in 3D space is known as the 3D Percentage of Correct Keypoints (3D-PCK), which considers a threshold in distance to decide if a joint is correctly predicted.

Moreover, the Area Under Curve (AUC) metric is also noticeably used alongside the 3D-PCK while evaluating model performance over varying distance threshold levels to

Table 4 Comparative analysis of 3D human pose datasets used in pose Estimation research

Dataset	Modalities	Key features	Subjects/actions	Use cases	Challenges addressed	Notable aspects
HumanEva-I	RGB, Grayscale video, 3D MoCap	Synchronized multi-view, 2D/3D keypoints	4 subjects, 6 actions	Sports analytics, robotics, AR/VR	Multi-view pose estimation	Calibrated videos, marker-based motion capture
Human3.6 M	RGB, Depth, 3D skeletons, IMU	4 M poses, 50 Hz capture, multi-view	11 subjects, 17 actions	Robotics, VR, biomechanics	Viewpoint generalization	Largest real-world dataset, synchronized multimodal data
Marconi	RGB (smartphone/GoPro), 3D poses (MoCap/A3D/DMC)	Multi-camera systems, varied environmental conditions	12 motion segments	Sports analysis, VR, autonomy	Cross-camera compatibility	Time-synchronized multi-view, three annotation methodologies
MPI-INF-3DHP	RGB, Markerless 3D poses	Casual clothing, chroma-key masks	8 subjects, 8 actions	AR/VR, real-world scenarios	Clothing variation, background noise	No markers, clothing diversity, background removal
Total Capture	Multi-view video, IMU, Vicon MoCap	Multi-modal, everyday clothing, background subtraction	5 activities (repeated)	Markerless motion capture, AR/VR	Occlusion, real-world dynamics	Combines IMU, video, and MoCap; foreground-background separation
SURREAL	Synthetic RGB, 3D poses, depth maps	SMPL models, CMU mocap, realistic textures	3.6 M frames, 145 subjects	Animation, gaming, AR	Synthetic-to-real generalization	Largest synthetic dataset, lighting/shadow simulation
Unite the People	RGB images, 3D body models	Dense landmarks (91 points), SMPLify optimization	6,777 images	Healthcare, biomechanics	Occlusion, body shape variation	Semi-automated labeling, regression tree for fast inference
JTA	Synthetic RGB (GTA V), 3D poses	Urban scenarios, occlusions, scale variation (0.1–100 m)	512 clips (pedestrians)	Surveillance, autonomous driving	Occlusion, scale/resolution changes	Realistic synthetic environments, cost-effective scalability
3DPW	RGB video, IMU data	Real-world motions, VIP method, 18 outfits	Daily activities	AR, practical pose estimation	Dynamic motions, clothing variation	IMU-sensor fusion (17 sensors/subject), real-world annotation
CMU Panoptic	480 VGA, 30 HD, 10 RGB-D sensors	Multi-person interactions, 521 synchronized cameras	Group activities	Robotics, surveillance, social VR	Multi-person tracking, crowd dynamics	Largest multi-camera setup (521 cameras), social behavior analysis

provide further accuracy. Choosing the right evaluation metric matters tremendously when trying to perform effective benchmarking of pose estimation models. For 2D pose estimation, metrics such as PCK, OKS, and mAP are used to assess the accuracy of key point localization. In contrast, NME is very helpful in tasks like facial landmark detection. In 3D pose estimation, the primary metrics for assessing the selection of spatial accuracy are the MPJPE, PA-MPJPE, and 3D-PCK. At the same time, for evaluating the smoothness of motion sequence in time, the MPJVE is the measure for the motion. Having these metrics allows the systematic analysis of various pose estimate techniques, facilitates model fine-tuning, and improves the functionality of pose estimation systems throughout their application spectrum, such as in augmented reality, healthcare, or human-computer interaction.

The summary comparative analysis of the most popular evaluation metrics of 2D and 3D pose estimation is compiled in Table 5, which presents their main objectives, benefits, cons, limitations, and scope of application.

4 Challenges and applications

This chapter builds on previous ones by detailing constraints that arise in the real world when using pose estimation techniques, including occlusion, ambiguity in-depth, and limits in computation. Furthermore, it reviews the advanced healthcare, surveillance, and AR/VR applications, illustrating how theoretical progress and datasets are further resourceful.

4.1 Challenges

The progress in human pose estimation so far is phenomenal due to the introduction of different modeling methodologies such as 2D kinematic, 2D planar, 3D kinematic, and volumetric 3D models. Regardless of the progress, all models, with their overlapping and unique challenges, suffer from some constraints impacting their accuracy and robustness under uncontrolled, real-world conditions, which are very difficult to work with. These are but are not limited to occlusion, depth ambiguity, computational complexity, lighting, shadows and reflections, and low light, to name a few. In addition, model-specific limitations like quantization errors from heat maps and rigid assumptions from planar models complicate accurate human pose estimation. The limits for each model pose are as follows:

4.1.1 2D kinematic models

The 2D kinematic models utilize pose estimation algorithms to get people's skeletons by estimating the joints' positions on a given 2D image. This is usually done using heatmaps, which are inferred from CNNs. The biggest issues of these models involve occlusions, diverse appearance of people, complex backgrounds, and changes in lighting conditions, such as shadows, reflections, and low light. Changes in lighting greatly impact pixel intensities and image gradients, making it impossible to identify the key points' location precisely. Also, depth perception is ambiguous because only planar coordinates are predicted without any direct 3D information. Additionally, the discretization of heatmaps leads to quantization errors, which reduces the precision in locating the key points. Under these circumstances, positional estimation of small and occluded

Table 5 Comparative analysis of 2D and 3D pose Estimation evaluation metrics

Metric	Dimensions	Description	Strengths	Weaknesses	Applications
PCK (Percentage of Correct Keypoints)	2D	Proportion of correctly predicted keypoints within a normalized distance	Scale-invariant, common benchmark	Threshold-dependent, less sensitive to minor errors	2D pose estimation, activity recognition, sports analysis
PCKh (Head-Normalized PCK)	2D	PCK with head size normalization	Robust to body size variations	Relies on accurate head size estimation	Multi-person tracking, MPII/LSP datasets
OKS (Object Keypoint Similarity)	2D	Gaussian-based similarity, COCO standard	Accounts for varying Keypoints difficulty	Computationally intensive, requires per-keypoint weights	COCO challenge, object pose estimation
mAP (Mean Average Precision)	2D	Average precision across OKS thresholds	Evaluates precision-recall trade-off, robust	Sensitive to small data changes, threshold dependent	Large-scale 2D pose estimation, COCO benchmark
NME (Normalized Mean Error)	2D	Normalized Euclidean distance between predicted and actual Keypoints	Scale-independent, good for landmarks	Ignores occlusions/missing points	Facial/hand landmark detection
MPJPE (Mean Per Joint Position Error)	3D	Average Euclidean distance between predicted and actual 3D joints	Direct spatial accuracy measure.	Sensitive to global misalignment	3D pose estimation, motion capture
PA-MPJPE (Procrustes-Aligned MPJPE)	3D	MPJPE after rigid alignment	Removes global misalignment, better structural evaluation	Less useful for absolute position accuracy	Cross-dataset 3D comparisons, 3D pose tracking
MPJVE (Mean Per Joint Velocity Error)	3D	Velocity error between predicted and actual joint movements	Evaluates temporal consistency	Not for single-frame predictions	Motion capture, activity recognition, video-based pose
3D-PCK (3D Percentage of Correct Keypoints)	3D	Proportion of correct 3D Keypoints within a distance threshold	Threshold-based, robustness analysis	Not continuous accuracy measurement	3D action recognition, occlusion handling
AUC (Area Under Curve)	3D	Accuracy aggregated over 3D-PCK thresholds	Holistic performance assessment	Computationally expensive	3D motion modeling, robotics

joints becomes very difficult. All these factors contribute to inaccurate estimation of the skeleton, especially at small or obscured joints.

On the other hand, methods based on regression are equally as complicated because of their tendency to oversimplify the spatial relationship, resulting in biomechanically unreasonable poses. Most 2D kinematic models worked to some degree in controlled settings but did not consider the changing lighting. Most approaches neglect this brutal case. Furthermore, many datasets captured for training and evaluation purposes are studio-lit, highlighting the lack of variability. The absence of explicit normalization photometric procedures or augmentation for illumination is a limiting factor contributing to the model's robustness to shadows, glare, or night imagery. Additionally, the quantization of heat maps is often underemphasized. Yet, it can significantly impact the accuracy of estimations near the joint and small limb boundaries where lighting artefacts further distort the heatmap. While regression models may offer more flexibility, they lack adjustable frameworks designed to counter lighting noise and perform poorly

under such conditions. This inadequate robustness limits the deployment of models in uncontrolled outdoor or industrial environments.

4.1.2 2D planar models

Planar human pose estimation models rely on geometry to define human limbs and their connectivity to other parts within the image plane, assuming it is 2D. A primary concern of these models is their reliance on clear outlines of the limb edges, which is heavily dependent on lighting. Shadows, highlights, and reflections can hide some boundaries, resulting in incorrect body structure estimation. In outdoor and/or cluttered spaces, specular reflection and non-uniform illumination lead to segmentation errors which are cascaded through planar pose estimation algorithms. In addition, these models operate on the assumption that limbs are rigid, planar mechanisms, which is a far too simplistic view capturing the phenomenon of human articulation. This assumption becomes especially problematic under challenging lighting conditions, where limb parts may blend into the background or overlap with one another, further reducing model accuracy. Most inadequately defined planar models do not incorporate enough mechanisms to deal with distortions due to lighting, which can lead to unreliable detection of limbs in uncontrolled environments. Evaluation datasets are created assuming uniform illumination, which dramatically narrows representativeness. Edge detection and segmentation techniques are known to be not very robust to photometric noise, but normalization of illumination or shadow removal is seldom applied to planar pose pipelines. Therefore, lighting noise is frequently added to pre-existing errors: discretization on the capturing of angles of limbs and the regression of orientations of segments. Failure to deal robustly with illumination variability means planar models are impractical for surveillance or sports analysis, where lighting changes quickly. The lack of investigations focusing on the combined effects of lighting artefacts and geometric simplifications indicates that these models lack a practical robustness approach, posing an important gap in research.

4.1.3 3D kinematic models

Multiview 3D kinematic models attempt to integrate detections from several 2D joints captured using single or multiple RGB cameras while estimating a human pose in 3D space. One of the issues is accurate joint position identification across different views with varying lighting conditions, which complicates the triangulation process. Joints can be hidden by shadows or inconsistent illumination from different cameras, and at the same time, false features can be added, increasing the complexity of proving definitive triangulation. Within a sequence of frames, joint pose tracking can be destabilized by temporal changes in the lighting due to moving light sources or environmental changes. Furthermore, intrinsic depth ambiguity, especially with monocular inputs, is a problem for 3D pose estimation. Reliable estimation is complicated by considerable position searching within discretized 3D space and the necessity for extensive pose data, including translations and rotations.Current 3D kinematic techniques still consider the effect of lighting discrepancy across views minimal. In essence, these methods degrade performance in real-world scenarios. Noise caused by photometric differences between cameras becomes an issue that spreads through multiview fusion and temporal smoothing stages; however, few methods utilize adaptive illumination feature extraction, illumination-invariant feature extraction, or active lighting compensation.

As a result, pose representation in 3D space is complex, and lighting features are poorly defined. Here, joint reconstructions are highly unstable and worsen. The biasness in the datasets captured in well-lit indoor environments limits the generalization to natural environments with other illumination dynamics. Also, little to no attention has been given to exploring the impact of light illumination on heatmap quantization and regression errors in the 3D spatial context, which is highly critical for robustness. Such limitations underline the need for specific models that explicitly integrate photometric modeling with data augmentation and are aware of the illumination.

4.1.4 3D volumetric models

Multi-view RGB representations provided as inputs to a human-dedicated 3D volumetric model serve to voxel the grid and volumetrically reconstruct the dense geometry of a human. One of the remaining issues for these models is the need to address the inconsistencies in lighting, which result in color and reflectance variations over time and across different views. Shadowing effects caused by limbs and neighbouring structures lead to partial occlusions and discontinuities in the voxel's occupancy. At the same time, low light conditions worsen the image quality, and the reliability of volumetric fusion is further degraded. Strongly non-Lambertian surfaces and specular reflection violate the diffuse reflectance assumptions that nearly all volumetric reconstruction relies on, resulting in incomplete or noisy shapes. Also, the real-time requirements of volumetric fusion's computational demands limit the use of advanced photometric corrections.

Research on volumetric pose estimation tends to ignore the more general case of lighting variability during voxel integration as the photometric conditions are ideal or controlled. This poses a problem in reality, where shadows and other lighting transitions tend to occur. The illumination variability degrades the pose's fidelity and anatomical realism by introducing erroneous occluded voxels and misleading surface reconstructions. The available techniques for volumetric fusion do not sufficiently capture features that are invariant to illumination or use illumination-variance normalization, and do not employ modern photometric noise with adequate filtering, which is also scribed dynamically. Also, there is a lack of financial backing for adopting illumination-conscious models in computationally expensive frameworks, which are widely used, especially for dynamic scenes. There is a need to develop means to deal with these issues in the balance of photometric strength and efficacy.

A comparative overview of the leading human pose estimation models, such as 2D kinematic, 2D pannar, 3D kinematic, and 3D volumetric, has been outlined in Table 6. The comparison analyzes the approach taken by each model, the methods of data collection and processing, and the challenges each faces. Critical limitations are also discussed, including quantization error in heatmap methods, depth ambiguity in 3D models, occlusion, lighting, and computational complexity. The table offers a concise summary aiding the understanding of what each model does and the practical trade-offs in different scenarios.

Cameras as sensors still suffer from occlusion and illumination change issues for human activity recognition and pose estimation. Due to dynamic conditions and resource constraints, the real-world context adds even further difficulty. Multi-modal data integration, simulation-based training paradigms, architectural changes, and other novel methods focus on solving the described problems. Partial visibility and lighting

Table 6 Comparison of 2D and 3D human pose estimation models across key technical aspects

Model type	Challenges	Critical analysis	Data dependency	Computational demand	Generalization ability
2D Kinematic	Occlusion, background clutter, Viewpoint dependency, Lack of depth, Motion blur in videos	Heatmap-based methods suffer from quantization error during keypoint extraction, Regression-based models lack spatial reasoning, Lighting variations severely affect detection accuracy due to shadow/highlights	Requires large labeled 2D datasets (e.g., COCO, MPII)	Low to moderate (efficient real-time models exist)	Good in controlled environments; weak in the wild
2D Planar	In-plane rotation, Simplified assumptions about human anatomy, Poor handling of self-occlusion and non-frontal views	Inflexible to complex articulation due to rigid shape templates, Prone to errors under illumination changes, Weak depth intuition limits interaction with 3D environments	Moderate; template-based training reduces diversity need	Low to moderate	Poor generalization due to limited representation power
3D Kinematic	Depth ambiguity from 2D lifting, Large search space for joints in 3D, Occlusions and inter-joint interference, Dependence on camera calibration or tracking	3D regression suffers from lack of geometric constraints, Heatmap quantization more severe in 3D, Lighting variation causes inconsistency in temporal reconstruction	High; requires multi-view or motion-capture datasets	High (especially for multi-view temporal models)	Moderate; improves with temporal or multiview cues
3D Volumetric	High computational and memory cost, Requires dense annotations, Poor performance in unconstrained scenes, Sensitive to occlusions, multiple people, and cluttered backgrounds.	Prone to resolution-computation trade-off in voxel space, Training requires expensive ground truth, Illumination inconsistency disrupts volumetric alignment and appearance modeling	Very high; limited public datasets (e.g., Human3.6 M, Panoptic)	Very high (due to 3D convolutions and voxel processing)	Weak in the wild unless trained with domain adaptation

inconsistency are being reduced through generative models with attention mechanisms and sensor fusion. The following works show different approaches researchers have taken to solve these issues across different contexts and modalities.

The approach presented by Han et al. [119] mitigates occlusion impacts for human pose estimation by employing the occlusion simulation tactic during training and applying dynamic structure loss to model the spatial interaction of joints. This further enhanced the prediction of key points during occlusion. Dutta et al.'s approach [120] also attempts to alleviate occlusion in human pose estimation by using occlusion simulation during training and dynamic structure loss to model the joints' spatial relationships, thus improving the prediction of occluded key points. None of the variations due to illumination changes are tackled, given the focus on joint configurations. Jiang and Xia [121] introduced a two-channel network for human pose estimation with occlusion reasoning. An OHE attention mechanism is developed specifically for occlusion problem-solving to focus on uncovered key points and boost feature representation for better joint estimation under occlusion.

Simultaneous processing of visible and occluded key points uses the dual-channel design, which enhances contextual comprehension and spatial reasoning. In extensive

evaluations, the proposed method outperforms existing approaches in the accuracy of pose estimation accuracies with occlusion. Ai et al. [122] propose a dual-teacher-based 2D human pose estimation approach in extremely low-light conditions with two antagonistic teachers generating reliable pseudo-labels on unlabeled data. This strategy minimizes direct and indirect illumination and occlusion, overcoming accuracy challenges in low-light benchmarks. Salman et al. [123] improve 2D hand pose estimation by incorporating ConvNeXt and contextual representations into a compact CPM that enhances robust lighting variations during feature extraction. Mathe et al. [73] introduced a new data augmentation technique for HAR where occlusions are added to the skeletal representation during training. This strategy mitigates the occlusion issues by enhancing partial obscuring scenarios that normally deteriorate the model's performance. Unlike traditional augmentation approaches focusing on transformations such as rotation, this method aims to address the occlusion problem directly, which may improve HAR performance in dynamic environments.

Wu et al. [124] present a new comprehensive two-phase architecture for human skeleton action recognition using Generative Adversarial Imputation Nets (GAIN) and a masked autoencoder. This approach seeks to resolve problems such as target occlusion and action similarity by utilizing GAIN to supplant the missing joint information, then applying a masked autoencoder to feature spatial-temporally reconstruct and learn. The framework already works better than others in recognizing actions when occluded or when action similarity constraints are applied. The method also uses data more readily than other methods, proving advanced generalization capabilities and aiding its applicability in practical scenarios where data is sparse or vague. Feng et al. [57] present CMDN method for the HAR framework that integrates multi-domain radar data for enhanced accuracy and robustness. With feature integration from various radar modalities, CMDN captures the complex dynamics of human activity, overcoming occlusion and varying illumination challenges. The approach is now more effective at recognizing activities performed in diverse environmental settings, demonstrating its applicability in real-world situations.

As described in [65], the problem of occlusion is handled through robust 2D pose estimation and the application of SLBT and hierarchical skeleton features that provide the correct joint-level information even when some portions of the body are covered. This blend of features allows the model to resolve posture-related ambiguities due to occlusions, which helps accurately detect shoplifting. An example is in [74], where Meng et al. described a framework for indoor activity recognition using multiple dynamic nonlinear mappings on 3D LiDAR data. Depth-based LiDAR data is less prone to occlusion and illumination changes, increasing accuracy when performing indoor activity recognition.

Other studies recently focused on using generative AI for pose estimation improvement under more difficult conditions. An example is Wang and Mao [125], who proposed an RFID sensing technique for augmenting and completing 3D human poses using generative AI that can, to some extent, alleviate occlusions by predicting obstructed joints without vision. Liu et al. [126] published work on EITNet. This IoT-based real-time basketball action recognition system uses sensor fusion to mitigate the effects of visual occlusions and changes in ambient lighting.

The described method shows the promise of combining IoT data for pose estimation algorithms functioning with 2D models within extremely intricate and lively settings.

Gang and Wang [127] provided an all-encompassing survey on advanced techniques of predicting, reconstructing, and generating human motion, paying particular attention to the application of generative models for pose estimation that include reconstruction of missing or occluded joints. Their work has noted that increasing attention is being given to modeling done within partial visibility or difficult illumination conditions to strengthen robustness. Fu et al. [128] suggested a hybrid architecture, namely HDiffTG, that integrates diffusion models, transformers, and graph convolutional networks for 3D human pose estimation. Although their focus is mainly on the 3D dimension, their method captures the streaks of contextual cues and long-range dependencies, which help concealment, portraying greater strength to occlusion. Such tactics are increasingly being employed in sophisticated 2D planar models to improve operational efficiency in visually demanding situations. Wang and Mao [125] presented a generative AI design based on RFID sensing interfaces for augmenting and completing 3D human poses. Their approach deals with occlusion problems and utilizes a latent diffusion transformer (LDT) with cross-attention conditioning. It facilitates the reproduction of 25-joint skeletal poses from partial RFID input data that usually contains only 12 joints. This provides a better estimation of the pose in the presence of occlusions or low visibility, which improves the performance of continuous 3D human pose estimation systems with wireless sensing technologies. Xiao et al. [129] propose a contrastive learning framework (CLAR) for human activity recognition (HAR) based on WiFi Channel State Information (CSI) using a diffusion model. To deal with issues like occlusion or sparse training data, these authors developed a denoising diffusion probabilistic model (DDPM) specific for time-series data augmentation. This model splits conditions into high-frequency and low-frequency components and applies them to generative processes using different weights to render quality augmentations. Moreover, an adaptive weight approach enhances the strength of positive sample pairs in contrastive learning to improve representation learning. All these changes together increase the robustness of the model to occlusions and changes to normal variability in human behavior. Chen et al. [118] describe a 3D human pose estimation diffusion model that predicts hidden joints based on observed data, thus allowing accurate full pose reconstruction even when some body parts are not fully visible.

Table 7 compares recent approaches addressing occlusion and illumination challenges in human pose estimation and activity recognition. This table outlines each study's methodology and key contributions used to enhance model robustness under visual constraints.

4.2 Applications

A few applications of pose estimation include movement analysis in human motion, where the movement is monitored for performance and feedback. It is also widely used in behavior recognition, which helps define the actions and interactions of people within different sets. Other notable applications are in gesture-based systems that make it possible to communicate with machines through body movements. Anomaly detection benefits from pose estimation from monitoring abnormal movements within specific regions. Moreover, it is significant in automated motion synthesis of realistic human actions in virtual environments. As the subsequent sections explain, the applications

Table 7 Analysis of state-of-the-art methods tackling occlusion and lighting challenges in human posture Estimation

Study	Approach/technique	Key contribution/outcome
Han et al. [119]	Occlusion simulation, dynamic structure loss	Improved keypoint prediction under occlusion
Dutta et al. [120]	Occlusion simulation, dynamic structure loss	Improved prediction of occluded keypoints
Jiang and Xia [121]	Two-channel network, OHE attention	Improved joint estimation under occlusion
Ai et al. [122]	Dual-teacher learning, pseudo-labels	Handles occlusion and low-light conditions
Salman et al. [122]	ConvNeXt + CPM	Improved robustness to lighting variations
Mathe et al. [73]	Occlusion-based augmentation	Enhanced HAR under partial occlusion
Wu et al. [124]	GAIN + masked autoencoder	Improved recognition under occlusion and action similarity
Feng et al. [57]	CMDN, multi-domain radar data	Robust to occlusion and illumination variations
Zhang and Wang [65]	SLBT + hierarchical skeleton features	Resolves occlusion for posture-based activity recognition
Meng et al. [74]	3D LiDAR, nonlinear mappings	High accuracy under occlusion and illumination change
Wang and Mao [125]	RFID + generative AI	Predicts occluded joints without vision
Liu et al. [126]	IoT-based sensor fusion (EITNet)	Mitigates occlusion and lighting changes
Gang and Wang [127]	Survey on generative models	Focus on occluded/missing joint reconstruction
Fu et al. [128]	HDiffTG (diffusion + GCN + transformer)	Robust to occlusion using contextual cues
Wang and Mao [125]	RFID + LDT with cross-attention	Reconstructs full pose from partial data
Xiao et al. [129]	CLAR with DDPM	Augments data under occlusion/sparse conditions
Chen et al. [118]	Diffusion model	Predicts hidden joints for full pose recovery

discussed above show how pose estimation enhances interaction, monitoring, and automation across many fields.

a. *Activity identification, anticipation, categorization, and monitoring*: Pose metadata has been employed as signals in many algorithms, such as action identification, prediction, recognition, and tracking [130]. suggested an actual action identification system based on pose-implemented algorithms. Added to action recognition technology. Studies [131] & [132] examined human location graphs to recognize distinct human activities in recordings. Research [37] utilized his calculated three-dimensional location to forecast long-term human migration. Method [38] suggested a perspective Bayesian posture embedding for video orientation. Posture-based video monitoring preserves people's privacy by concentrating on their online posture and depiction rather than personal information.
b. *Action correction*: Certain sports, such as dancing, track and field, and professional training, need careful control of the human body to preserve optimal posture. Personal trainers generally manage posture modifications and activity courses face-to-face. 3D HPE and action recognition enable AI personal trainers to set up cameras and assist courses without a personal trainer present. An AI coaching system [39] designed for individual sports training assistance was integrated with a posture estimate module.
c. *Animation, movie & gaming*: Many performance capture firms in animation, film, and games employ motion capture to show characters with exact motions and authentic physical processes. Motion capture equipment is generally costly and complicated to set up. Human Pose Estimation can offer precise location data without requiring pricey specialist equipment [133, 134].

d. *AR and VR*: Immersive Virtual reality technology strives to enhance the interaction between digital things and the actual world. The purpose of fully interactive technology is to immerse individuals in immersive experiences. Augmented and virtual reality technologies employ human positional information in addition to meeting the aims of many applications. Animated characters may be made to represent actual individuals in genuine circumstances. Wen et al. [135] integrated three-dimensional position prediction with human mesh reconstruction to produce three-dimensional computer animators from a single picture [136]. proposed a pose-based approach for transforming footage of aired tennis tournaments into interactive and controlled visual sprites.
e. *Healthcare*: HPE supplies physicians with the measurable motion data needed to perform rehab, build up treatment regimens, and identify challenging medical disorders. Research [137] presented a posture-based estimation technique for quantifying motor severity in Parkinson's disease. A study [138] created a posture-based physical therapy system that permits patient examination and counselling at home. Systems of this nature may also be designed to recognize aberrant behavior and anticipate future conduct. Alerts are delivered rapidly when the system finds a probable hazard [139]. gave prompt help leveraging HPE's fall detection monitoring technology.

5 Conclusions and future scope

This concluding chapter integrates information from previous chapters with subsequently identified gaps and sets out optimal strategy frameworks for further research. It highlights the paper's main contributions while suggesting assertive efforts in overcoming pose estimation challenges toward developing more reliable and efficient solutions adaptable to any setting.

This article thoroughly reviews the progress made in human pose estimation in both 2D and 3D reconstruction, emphasizing the state-of-the-art techniques utilized, the difficulties involved, and what the future might hold. Large-scale datasets coupled with DL techniques have the most impact on improving the accuracy of pose estimation models as they can leverage DL approaches, which have advanced tremendously today. While there is substantial improvement in model predictions, occlusion in images, illumination changes, and background clutter still need to be accounted for in real-life situations. Moreover, although the provided datasets have many features, the annotated 3D data is insufficient to allow models to perform well in diverse environments. To overcome these obstacles, there is a need for more comprehensive training approaches, better domain adaptation and self-supervised learning strategies that would enhance the efficacy of the models and their generalization. However, the development of the techniques opens the opportunity to explore more issues revolving around the reliability, trust, and federated learning concerning the 2D and 3D human pose estimation, which indeed is aimed to be solved in the upcoming years, especially with the application of novel DL techniques in the area.

Unsupervised learning Currently, fully supervised algorithms dominate human pose estimation due to their strong performance. The success relies on the quantity of pose annotations in your big dataset. On the other hand, images and videos without captions remain

accessible nearly eternally, making it hard to annotate this asset adequately. Therefore, significant progress has been achieved in unsupervised learning that can automatically extract information about the human body from large volumes of data.

Attitude display The heatmap-based attitude display was superior to the others. However, the quantization error is inherent when encoding and decoding heat maps from coordinates. Resolution impacts both heat map encoding and decoding. The greater the resolution, the better the accuracy, but the more computing work. Therefore, strong bodily indifference is essential to cope with such situations.

Model explainability: Lack of interpretation is a shortcoming of DL systems. A full and formal theory of interpretability does not yet exist. Therefore, there is no systematic direction in creating DL models. Human posture estimates also lack a clear knowledge of how visual features of the input picture impact final key point localization, damaging future studies. Due to their evident inadequacies, we highly urge ongoing study into the interpretability of human pose estimate algorithms. Multi-Person Her three-dimensional objects Pose Estimation: In actuality, multi-person settings are prevalent.

Model-based method Using a model-based concept offers several advantages. B. Stability and Scalability. Regarding stability, the model-based technique surpasses the skeleton-based strategy, notably under occlusion and in the wild. A model-based solution can be readily introduced.

- More critical point information is available for more powerful monitoring.
- Interpenetration errors can be reduced by taking advantage of scene limits and multi-person situations.
- Spatio-temporal information can be used in a simple way. Only pose and body type parameters should be considered.
- It can bridge the gap between human pose and appearance structure.

The most recent breakthroughs in DL make it possible to reconstruct object geometry accurately using only RGB images. This helps in 3D pose estimation since sparse skeletal contact points and dense mesh full-body shape representation can be captured simultaneously, leading to a better understanding of human motion. In addition, the accuracy of occlusion handling and depth estimation improves with the integration of object geometry reconstruction for 3D human localization. Regardless, the combined study of scene comprehension, human-object relationships, and environment rebuilding is still novel and lacks comprehensive study. Closing this gap can construct more reliable pose estimation models that interpret complex environments.

In the presence of humans within an image, the scene's context is understood much better; thus, pose estimation can be used to extract information from the image. Examination of pose-based attributes, like body position, motion intent, and object interaction, helps to understand a given situation better. However, most current approaches concentrate on action recognition and anomaly detection instead of scene comprehension based on movement and behavioural pattern recognition. Further work may enable pose estimation to extend beyond individual action recognition with the aid of scene-aware perception models, which reason about human intent and context for the environment, to enhance autonomous systems, surveillance, and assistive robotics.

Another important future direction is improving pose estimation's speed and computation cost to mobile devices, edge computing, and embedded systems. One of the most important aspects of model accuracy in augmented reality, robotics, autonomous driving, and real-time motion analysis is maintaining model accuracy while decreasing model complexity. Moreover, integrating pose estimation for human subjects with animal motion analysis, sign language interpretation, and medical image evaluation broadens the scope of research possibilities. Ethics and equity in the automation of pose estimation processes also need to be studied so that different demographic groups are not disadvantaged because of biased pose estimation models. Solving these problems will enhance the adaptation and availability of pose estimation models to other fields of study and industry.

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References

1. Muzammul M, Li X. Comprehensive review of deep learning-based tiny object detection: challenges, strategies, and future directions. Knowl Inf Syst. 67. 2025;3825–3913. https://doi.org/10.1007/s10115-025-02375-9
2. Ji L, Du Y, Dang Y, Gao W, Zhang H. A survey of methods for addressing the challenges of referring image segmentation. Neurocomputing. 2024;583:127599.
3. Liu W, Chen J, Li C, Qian C, Chu X, Hu X. A cascaded inception of inception network with attention modulated feature fusion for human pose estimation. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 32, No. 1). 2018.
4. Tang W, Yu P, Wu Y. Deeply learned compositional models for human pose estimation. In Proceedings of the European conference on computer vision (ECCV) 2018. pp. 190–206.
5. Tang W. Human pose estimation: single-person and multi-person approaches. In ITM Web of Conferences, Vol. 70. EDP Sciences; 2025. p. 02019.
6. Ansari MA, Singh DK. Identifying human activities in megastores through postural data to monitor shoplifting events. Neural Comput Appl. 2023;35(9):6515–28.
7. Fang HS, Li J, Tang H, Xu C, Zhu H, Xiu Y, Lu C. Alphapose: Whole-body regional multi-person pose Estimation and tracking in real-time. IEEE Trans Pattern Anal Mach Intell. 2022;45(6):7157–73.
8. Li W, Wang Z, Yin B, Peng Q, Du Y, Xiao T, Sun J. Rethinking on multi-stage networks for human pose estimation. 2019. arXiv preprint arXiv:1901.00148.
9. Kumar P, Chauhan S, Awasthi LK. Human activity recognition (har) using deep learning: review, methodologies, progress and future research directions. Arch Comput Methods Eng. 2024;31(1):179–219.
10. Khan SI, Dawood H, Khan MA, Issa GF, Hussain A, Alnfiai MM, Adnan KM. Transition-aware human activity recognition using an ensemble deep learning framework. Comput Hum Behav. 2025;162:108435.
11. Ansari MA, Singh DK. ESAR, an expert shoplifting activity recognition system. Cybern Inf Technol. 2022;22(1):190–200.
12. Zuffi S, Freifeld O, Black MJ. From pictorial structures to deformable structures. In 2012 IEEE conference on computer vision and pattern recognition. IEEE; 2012, pp. 3546–3553.
13. Zhang Y, You S, Karaoglu S, Gevers T. 3D human pose estimation and action recognition using Fisheye cameras: a survey and benchmark. Pattern Recognition; 2025. p. 111334.

14. Hossen MA, Abas PE. Machine learning for human activity recognition: State-of-the-Art techniques and emerging trends. J Imaging. 2025;11(3):91.
15. Yan L, Du Y. Exploring trends and clusters in human posture recognition research: an analysis using CiteSpace. Sensors. 2025;25(3):632.
16. Pons-Moll G, Romero J, Mahmood N, Black MJ. Dyna: A model of dynamic human shape in motion. ACM Trans Graphics (TOG). 2015;34(4):1–14.
17. Chen Y, Tian Y, He M. Monocular human pose estimation: A survey of deep learning-based methods. Comput Vis Image Underst. 2020;192:102897.
18. Mittal A, Zhao L, Davis LS. Human body pose estimation using silhouette shape analysis. In Proceedings of the IEEE conference on advanced video and signal based surveillance, 2003. IEEE;2003, pp. 263–270.
19. Reinbacher C, Ruether M, Bischof H. Pose estimation of known objects by efficient silhouette matching. In 2010 20th International Conference on Pattern Recognition. IEEE; 2010, pp. 1080–1083.
20. Dutta A, Lal R, Raychaudhuri DS, Ta CK, Roy-Chowdhury AK. Poise: Pose guided human silhouette extraction under occlusions. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision 2024. pp. 6153–6163.
21. Beddiar DR, Nini B, Sabokrou M, Hadid A. Vision-based human activity recognition: a survey. Multimedia Tools Appl. 2020;79(41):30509–55.
22. Liu C, Guo W, Hu W, Chen R, Liu J. Real-time model-based monocular pose tracking for an asteroid by contour fitting. IEEE Trans Aerosp Electron Syst. 2020;57(3):1538–61.
23. Wrench A, Balch-Tomes J. Beyond the edge: markerless pose Estimation of speech articulators from ultrasound and camera images using deeplabcut. Sensors. 2022;22(3):1133.
24. Kim TT, Zohdy MA, Barker MP. Applying pose Estimation to predict amateur golf swing performance using edge processing. IEEE Access. 2020;8:143769–76.
25. Erol A, Bebis G, Nicolescu M, Boyle RD, Twombly X. Vision-based hand pose estimation: A review. Comput Vis Image Underst. 2007;108(1–2):52–73.
26. Ansari MA, Kurchaniya D, Dixit M. A comprehensive analysis of image edge detection techniques. Int J Multimedia Ubiquitous Eng. 2017;12(11):1–12.
27. Ezzeldin M, Ghoneim A, Abdelhamid L, Atia A. Survey on multimodal complex human activity recognition. FCI-H Inf Bull. 2025;7(1):26–44.
28. Aldoma A, Marton ZC, Tombari F, Wohlkinger W, Potthast C, Zeisl B, et al. Tutorial: Point cloud library: three-dimensional object recognition and 6 dof pose estimation. IEEE Robot Autom Mag. 2012;19(3):80–91.
29. Titsias M. The infinite gamma-Poisson feature model. Adv Neural Inf Process Syst. 2007;20.
30. Ansari MA, Singh DK. Human detection techniques for real time surveillance: a comprehensive survey. Multimedia Tools Appl. 2021;80(6):8759–808.
31. Chen M, Quirk C. Embedding edge-attributed relational hierarchies. In Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval. 2019. pp. 873–876.
32. Wu J, Cui Z, Sheng VS, Zhao P, Su D, Gong S. A comparative study of SIFT and its variants. Meas Sci Rev. 2013;13(3):122.
33. Wu Q, Zhu P, Chai Z, Guo G. Joint learning of foreground, background and edge for salient object detection. Computer Vision and Image Understanding. 2024; 240:103915.
34. Pant N, Rodriguez IF, Beniwal A, Warren S, Serre T. HMAX strikes back: Self-supervised learning of human-like scale invariant representations. Cognitive Computational Neuroscience. 2024.
35. Naz H, Zhang Z, Al-Habib M, Awwad FA, Ismail EA, Khan ZA. Posture detection of heart disease UsingMulti-Head attention vision hybrid (MHAVH) model. Computers, Materials & Continua. 2024;79(2):2673–2696.
36. Lan G, Wei J, Huang H, Zou F, Li D. MSB-Net: an end-to-end network for extracting building from high-resolution remote sensing imagery. IEEE J Sel Top Appl Earth Observ Remote Sens. 2024. 17:6253–6264.
37. Cao Z, Gao H, Mangalam K, Cai QZ, Vo M, Malik J. Long-term human motion prediction with scene context. In Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part I 16. Springer International Publishing; 2020. pp. 387–404.
38. Sun JJ, Zhao J, Chen LC, Schroff F, Adam H, Liu T. View-invariant probabilistic embedding for human pose. In Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part V 16. Springer International Publishing; 2020. pp. 53–70.
39. Wang J, Qiu K, Peng H, Fu J, Zhu J. Ai coach: Deep human pose estimation and analysis for personalized athletic training assistance. In Proceedings of the 27th ACM international conference on multimedia, 2019. pp. 374–382.
40. Zang Y, Li W, Han J, Zhou K, Loy CC. Contextual object detection with multimodal large Language models. Int J Comput Vis. 2024;133(2):825–843.
41. Roggio F, Trovato B, Sortino M, Musumeci G. A comprehensive analysis of the machine learning pose estimation models used in human movement and posture analyses: a narrative review. Heliyon; 2024.
42. Gao Z, Chen J, Liu Y, Jin Y, Tian D. A systematic survey on human pose estimation: upstream and downstream tasks, approaches, lightweight models, and prospects. Artif Intell Rev. 2025;58(3):68.
43. Thu NV, Tri DQ, Hoa BTK, Nguyen-Thi HA, Hung NV, Tuyet QTT, et al. Application of optical flow technique to short-term rainfall forecast for some synoptic patterns in Vietnam. Theoretical Appl Climatol. 2025;156(1):1–29.
44. Stein A, Hebert M. Combining local appearance and motion cues for occlusion boundary detection. 2007.
45. Zhan SZ, Chang IC. Pictorial structures model based human interaction recognition. In 2014 International Conference on Machine Learning and Cybernetics, Vol. 2. IEEE; 2014, pp. 862–866.
46. Zhang X, Li C, Hu W, Tong X, Maybank S, Zhang Y. Human pose Estimation and tracking via parsing a tree structure based human model. IEEE Trans Syst Man Cybernetics: Syst. 2013;44(5):580–92.
47. Tai KS, Socher R, Manning CD. Improved semantic representations from tree-structured long short-term memory networks. ArXiv Preprint arXiv:150300075. 2015.
48. Fang HS, Cao J, Tai YW, Lu C. Pairwise body-part attention for recognizing human-object interactions. In Proceedings of the European conference on computer vision (ECCV), 2018. pp. 51–67.
49. Li S, Zhang J, Fu X, Zhang X, Shang J, Gupta R. Matching skeleton-based activity representations with heterogeneous signals for HAR. In Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems. 2025. pp. 574–587.

50. Yunardi RT, Sardjono TA, Mardiyanto R. Enhancing surveillance vision-based human action recognition using skeleton joint swing and angle feature and modified AlexNet-LSTM. Int J Intell Eng Syst. 2025;18(1).
51. Krishna SR, Pavithra M, Thatipudi JG, Rajkumar K. MNCL: empirical development of human action recognition based on postures using modified neural classification logic. In 2025 International Conference on Visual Analytics and Data Visualization (ICVADV). IEEE 2025. pp. 1299–1306.
52. Scanagatta M, Salmerón A, Stella F. A survey on bayesian network structure learning from data. Progress Artif Intell. 2019;8(4):425–39.
53. Pozo E, Gerges B, Nafea M, Schrödel F. Next level of Human-Robot collaboration by utilizing AI pose Estimation and model predictive motion planning technologies. IFAC-PapersOnLine. 2023;56(2):5313–9.
54. Salisu S, Mohamed ASA, Jaafar MH, Pauzi AS, Younis HA. A survey on deep learning-based 2D human pose estimation models. Comput Mater Continua. 2023;76(2).
55. Gong W, Zhang X, Gonzàlez J, Sobral A, Bouwmans T, Tu C, Zahzah EH. Human pose estimation from monocular images: a comprehensive survey. Sensors. 2016;16(12):1966.
56. Anguelov D, Srinivasan P, Koller D, Thrun S, Rodgers J. & SCAPE JD. Shape completion and animation of people. ACM Trans Graph. 24(3):408.
57. Feng X, Chen P, Weng Y, Zheng H. CMDN: continuous human activity recognition based on multi-domain radar data fusion. IEEE Sens J. 2025.
58. Wang J, Tan S, Zhen X, Xu S, Zheng F, He Z, Shao L. Deep 3D human pose estimation: A review. Comput Vis Image Underst. 2021;210:103225.
59. Elshami NE, Salah A, Mohsen H. A comparative study of recent 2D human pose estimation methods. In 2024 6th International Conference on Computing and Informatics (ICCI). IEEE;2024, pp. 528–537.
60. Khan MH, Shafiq H, Farid MS, Grzegorzek M. Encoding human activities using multimodal wearable sensory data. Expert Syst Appl. 2025.261:125564.
61. van der Waard MWP, de Jong LAF, Keijsers NLW. Motion tracking with automated pose estimator can enhance ankle-foot-orthoses alignment. Clin Biomech Elsevier Ltd. 2025;121:106375.
62. Zhao L, Lin Z, Sun R, Wang A. A review of State-of-the-Art methodologies and applications in action recognition. Electronics. 2024;13(23):4733.
63. Zhang Z, Luo Y, Gou J. Double anchor embedding for accurate multi-person 2D pose Estimation. Image Vis Comput. 2021;111:104198.
64. Cao Z, Hidalgo G, Simon T, Wei SE, Sheikh Y. Openpose: realtime multi-person 2d pose Estimation using part affinity fields. IEEE Trans Pattern Anal Mach Intell. 2019;43(1):172–86.
65. Zhang H, Wang F. Research on human action detection and recognition methods based on interpretable artificial intelligence. J COMBIN MATH COMBIN COMPUT. 2025;127:1143–58.
66. Ansari MA, Ahmad S, Mewada A, Maurya SK. PosturalNet, an advanced architecture to detect shoplifting at megastores. In 2025 IEEE 14th International Conference on Communication Systems and Network Technologies (CSNT). IEEE; 2025. pp. 863–867.
67. Nguyen T, Le V, Duong D, Pham T, Le D. 3D human pose Estimation in Vietnamese traditional martial Art videos. J Adv Eng Comput. 2019;3(3):471–91. https://doi.org/10.25073/jaec.201933.252.
68. Udayan J, Raman DTVJ, Kim R, H., Nedungadi P. Deep learning in monocular 3D human pose estimation: systematic review of contemporary techniques and applications. Multimed Tools Appl. 2025;1–37.
69. Guo Y, Gao T, Dong A, Jiang X, Zhu Z, Wang F. A survey of the state of the Art in monocular 3D human pose estimation: methods, benchmarks, and challenges. Sensors. 2025;25(8):2409.
70. Navone A, Martini M, Chiaberge M. Autonomous robotic pruning in orchards and vineyards: a review. 2025. arXiv preprint arXiv:2505.07318.
71. Zarkeshev A, Csiszár C. Rescue method based on V2X communication and human pose Estimation. Periodica Polytech Civil Eng. 2019;63(4):1139–46.
72. Rani M, Kumar M. MobileNet for human activity recognition in smart surveillance using transfer learning. Neural Comput Appl. 2025;37(5):3907–24.
73. Mathe E, Vernikos I, Spyrou E, Mylonas P. Leveraging artificial occluded samples for data augmentation in human activity recognition. Sensors. 2025;25(4):1163.
74. Meng X, Bouazizi M, Li Z, Ohtsuki T. Indoor human activity recognition using multiple dynamic nonlinear mapping applied to 3D LiDAR-Collected data. IEEE Internet Things J. 2025.
75. Xin C, Kim S, Cho Y, Park KS. Enhancing human action recognition with 3D skeleton data: a comprehensive study of deep learning and data augmentation. Electronics. 2024;13(4):747.
76. Sharma V, Gupta M, Kumar A, Mishra D. Star-3d: A holistic approach for human activity recognition in the classroom environment. Information. 2024;15(4):179.
77. Eder C. Pose estimation of deformable objects. Doctoral dissertation, Technische Universität Wien, 2024.
78. Sturm P. Algorithms for plane-based pose estimation. In Proceedings IEEE Conference on Computer Vision and Pattern Recognition, Vol. 1. IEEE; 2000, pp. 706–711.
79. Collins T, Bartoli A. Infinitesimal plane-based pose Estimation. Int J Comput Vision. 2014;109(3):252–86.
80. Qiu Z, Yang Q, Wang J, Wang X, Xu C, Fu D et al. Learning structure-guided diffusion model for 2d human pose estimation. 2023. arXiv preprint arXiv:2306.17074.
81. Lv X, Hao W, Tian L, Han J, Chen Y, Cai Z. LiteDEKR: End-to-end lite 2D human pose Estimation network. IET Image Proc. 2023;17(12):3392–400.
82. Lu T, Cheng K, Hua X, Qin S. KSL-POSE: A Real-Time 2D human pose Estimation method based on modified YOLOv8-Pose framework. Sensors. 2024;24(19):6249.
83. Li H, Shi B, Dai W, Zheng H, Wang B, Sun Y, et al. Pose-oriented transformer with uncertainty-guided refinement for 2d-to-3d human pose estimation. In Proceedings of the AAAI conference on artificial intelligence. 2023. 37(1):1296–1304.
84. Davoodnia V, Ghorbani S, Messier A, Etemad A. SkelFormer: markerless 3D pose and shape estimation using skeletal transformers. 2024. ArXiv Preprint arXiv:240412625.
85. Wu X, Ling Z, Zhang X, Ma Z, Deng W. Human similar activity recognition using Millimeter-Wave radar based on CNN-BiLSTM and class activation mapping. Eng. 2025;6(3):44.

86. Zhou Z, Shuai Q, Wang Y, Fang Q, Ji X, Li F, et al. Quickpose: real-time multi-view multi-person pose estimation in crowded scenes. In ACM SIGGRAPH 2022 Conference Proceedings, 2022. pp. 1–9.
87. Bogo F, Kanazawa A, Lassner C, Gehler P, Romero J, Black MJ. Keep it SMPL: Autormatic estimation of 3D human pose and shape from a single image. In European Conference on Computer Vision (ECCV). 2016.
88. Chen Z, Wang F, Wang Y, Liu H. Text-to-3d using gaussian splatting. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024, pp. 21401–21412.
89. Lassner C, Romero J, Kiefel M, Bogo F, Black MJ, Gehler PV. Unite the people: Closing the loop between 3d and 2d human representations. In Proceedings of the IEEE conference on computer vision and pattern recognition 2017, pp. 6050–6059.
90. Pavlakos G, Choutas V, Ghorbani N, Bolkart T, Osman AA, Tzionas D, Black MJ. Expressive body capture: 3d hands, face, and body from a single image. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2019, pp. 10975–10985.
91. Briceno L, Paul G. Makehuman: a review of the modelling framework. In Proceedings of the 20th Congress of the International Ergonomics Association (IEA 2018) Volume V: Human Simulation and Virtual Environments, Work With Computing Systems (WWCS), Process Control 20. Springer International Publishing; 2019, pp. 224–232.
92. Caudron R, Nicq PA. Blender 3D by example. Packt Publishing Ltd.; 2015.
93. Qammaz A, Argyros AA. MocapNET: ensemble of SNN encoders for 3D human pose estimation in RGB images. In BMVC. 2019. p. 46.
94. Jayamohan M, Yuvaraj S. Iv3-MGRUA: a novel human action recognition features extraction using inception v3 and video behaviour prediction using modified gated recurrent units with attention mechanism model. SIViP. 2025;19(1):1–12.
95. Ray LSS, Rey VF, Zhou B, Suh S, Lukowicz P. Human attribute guided dynamic ground pressure profile transfer using 3D simulated pressure maps. Activity Behav Healthc Comput. 2025. p. 1.
96. Jin G, Wang J, Wang Y, Du H, Ma L, Song Q. Multi-dataset pose estimation based on the fusion of multiple models. In Proceedings of the 3rd International Conference on Computer, Artificial Intelligence and Control Engineering. 2024, pp. 220–226.
97. Schneider S, Vollmer R. Poses of people in art: A dataset for human pose Estimation in digital Art history. ACM J Comput Cult Herit. 2024;17(4):1–19.
98. Arunabha C. Generalized FLIC: Learning with misclassification for Binary Classifiers, Doctoral dissertation, University of Kansas, 2014.
99. Luo W, Xue J. Human pose estimation based on improved HRNet model. In 2023 IEEE 3rd International Conference on Computer Communication and Artificial Intelligence (CCAI). IEEE; 2023, pp. 153–157.
100. Papandreou G, Zhu T, Chen LC, Gidaris S, Tompson J, Murphy K. Personlab: Person pose estimation and instance segmentation with a bottom-up, part-based, geometric embedding model. In Proceedings of the European conference on computer vision (ECCV). 2018, pp. 269–286.
101. Jain S, Dash S, Deorari R. Object detection using coco dataset. In 2022 International Conference on Cyber Resilience (ICCR). IEEE; 2022, pp. 1–4.
102. Wu J, Zheng H, Zhao B, Li Y, Yan B, Liang R, et al. Ai challenger:A large-scale dataset for going deeper in image understanding. arXiv preprint arXiv:1711.06475. 2017.
103. Golda T, Kalb T, Schumann A, Beyerer J. Human pose estimation for real-world crowded scenarios. In 2019 16th IEEE international conference on advanced video and signal based surveillance (AVSS). IEEE;2019. pp. 1–8.
104. Iqbal U, Garbade M, Gall J. Pose for action-action for pose. In 2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017). IEEE; 2017. pp. 438–445.
105. Andriluka M, Iqbal U, Insafutdinov E, Pishchulin L, Milan A, Gall J, Schiele B. Posetrack: A benchmark for human pose estimation and tracking. In Proceedings of the IEEE conference on computer vision and pattern recognition 2018, pp. 5167–5176.
106. Lin W, Liu H, Liu S, Li Y, Xiong H, Qi G, Sebe N. Hieve: A large-scale benchmark for human-centric video analysis in complex events. Int J Comput Vision. 2023;131(11):2994–3018.
107. Zhang S-H et al. Pose2Seg: detection free human instance segmentation. 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 889–898. 2019. https://doi.org/10.1109/CVPR.2019.00098.
108. Sigal L, Black MJ. Humaneva: synchronized video and motion capture dataset for evaluation of articulated human motion. Brown Univertsity TR; 2006. 120(2).
109. Ionescu C, Papava D, Olaru V, Sminchisescu C. Human3. 6m: large scale datasets and predictive methods for 3d human sensing in natural environments. IEEE Trans Pattern Anal Mach Intell. 2013;36(7):1325–39.
110. Elhayek A, de Aguiar E, Jain A, Thompson J, Pishchulin L, Andriluka M, et al. Marconi—convnet-based marker-less motion capture in outdoor and indoor scenes. IEEE Trans Pattern Anal Mach Intell. 2016;39(3):501–514.
111. Li C, Lee GH. Generating multiple hypotheses for 3d human pose estimation with mixture density network. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2019. pp. 9887–9895.
112. Trumble M, Gilbert A, Malleson C, Hilton A, Collomosse J. Total capture: 3d human pose estimation fusing video and inertial sensors. In Proceedings of 28th British Machine Vision Conference, 2017. pp. 1–13.
113. Joshi I, Grimmer M, Rathgeb C, Busch C, Bremond F, Dantcheva A. Synthetic data in human analysis: A survey. IEEE Trans Pattern Anal Mach Intell. 2024;46(7):4957–76.
114. Fabbri M, Lanzi F, Calderara S, Palazzi A, Vezzani R, Cucchiara R.Learning to detect and track visible and occluded body joints in a virtual world. In Proceedings of the European conference on computer vision (ECCV), 2018. pp. 430–446.
115. Gargi N, Kim E, Liang S, Ma G. Monocular pose estimation for VR motion capture applications. Training, 1: 0.
116. Arnab A, Doersch C, Zisserman A. Exploiting temporal context for 3D human pose estimation in the wild. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2019. pp. 3395–3404.
117. Hallman TA, Robinson WD. Comparing multi-and single-scale species distribution and abundance models built with the boosted regression tree algorithm. Landscape Ecol. 2020;35:1161–74.
118. Chen Z, Dai J, Pan J. A conditional diffusion model for 3d human pose estimation. In 2024 4th International Conference on Consumer Electronics and Computer Engineering (ICCECE). IEEE; 2024. pp. 20–24.
119. Han G, Song C, Wang S, et al. Occluded human pose Estimation based on limb joint augmentation. Neural Comput Applic. 2025;37:1241–53. https://doi.org/10.1007/s00521-024-10676-3.

120. Dutta A, Bose S, Bachu S, Ta CK, Karydis K, Roy-Chowdhury AK. Unsupervised domain adaptation for occlusion resilient human pose estimation. 2025. arXiv preprint arXiv:2501.02773.
121. Jiang J, Xia N. A dual-channel network based on occlusion feature compensation for human pose Estimation. Image Vis Comput. 2024;151:105290.
122. Ai Y, Qi Y, Wang B, Cheng Y, Wang X, Tan RT. Domain-adaptive 2d human pose estimation via dual teachers in extremely low-light conditions. In European Conference on Computer Vision. Cham: Springer Nature Switzerland; 2024. pp. 221–239.
123. Salman SA, Zakir A, Takahashi H. Adopting ConvNeXt and contextual representation for enhanced feature integration in compact CPM for 2D hand pose estimation. In International Workshop on Advanced Imaging Technology (IWAIT). 2024 SPIE. 2024. 13164:367–372.
124. Wu S, Lu G, Han Z, Chen L. A robust two-stage framework for human skeleton action recognition with GAIN and masked autoencoder. Neurocomputing. 2025;623:129433.
125. Wang Z, Mao S. Generative AI-Empowered RFID sensing for 3D human pose augmentation and completion. IEEE Open J Commun Soc; 2025.
126. Liu J, Liu X, Qu M, Lyu T. EITNet: an IoT-enhanced framework for real-time basketball action recognition. Alexandria Eng J. 2025;110:567–78.
127. Gang C, Wang Y. Human motion prediction, reconstruction, and generation. arXiv preprint arXiv:2502.15956. 2025.
128. Fu Y, Huang C, Li J, Kong H, Tian Y, Li H, Zhang Z. HDiffTG: A lightweight hybrid diffusion-transformer-GCN architecture for 3D human pose Estimation. arXiv preprint arXiv:2505.04276. 2025.
129. Xiao C, Han Y, Yang W, Hou Y, Shi F, Chetty K. Diffusion model-based contrastive learning for human activity recognition. IEEE Internet Things J; 2024.
130. Dubey S, Dixit M. A comprehensive survey on human pose estimation approaches. Multimedia Syst. 2023;29(1):167–95.
131. Long JA, Lee J, Reuschke D. Activity graphs: Spatial graphs as a framework for quantifying individual mobility. J Geogr Syst. 2023;25(3):377–402.
132. Markovitz A, Sharir G, Friedman I, Zelnik-Manor L, Avidan S. Graph embedded pose clustering for anomaly detection. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2000. pp. 10539–10547.
133. Xiao B, Wu H, Wei Y. Simple baselines for human pose estimation and tracking. In Proceedings of the European conference on computer vision (ECCV), 2018, pp. 466–481.
134. Liu J, Fu H, Tai CL. Posetween: Pose-driven tween animation. In Proceedings of the 33rd annual acm symposium on user interface software and technology, 2020. pp. 791–804.
135. De Aguiar E, Theobalt C, Stoll C, Seidel HP. Marker-less deformable mesh tracking for human shape and motion capture. In 2007 IEEE conference on computer vision and pattern recognition. IEEE; 2007. pp. 1–8.
136. Zhang H, Sciutto C, Agrawala M, Fatahalian K. Vid2player: controllable video sprites that behave and appear like professional tennis players. ACM Trans Graphics (TOG). 2021.40(3):1–16.
137. Martínez-Martín P, Rodríguez-Blázquez C, Alvarez M, Arakaki T, Arillo VC, Chaná P, et al. Parkinson's disease severity levels and MDS-Unified Parkinson's Disease Rating Scale. Parkinson Relat Disord. 2015;21(1):50–54.
138. Gu Y, Pandit S, Saraee E, Nordahl T, Ellis T, Betke M. Home-based physical therapy with an interactive computer vision system. In Proceedings of the IEEE/CVF international conference on computer vision workshops, 2019. pp. 0–0.
139. Lu M, Poston K, Pfefferbaum A, Sullivan EV, Fei-Fei L, Pohl KM et al. Vision-based estimation of MDS-UPDRS gait scores for assessing Parkinson's disease motor severity. In Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part III 23. Springer International Publishing;2020. pp. 637–647.

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